



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

19-03-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	B	D	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,D	A,B,C	A,C,D	A,B,C,D	B,D	D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.30	2.70	2.00	1.33	1.33	2.66 to 2.67
SECTION-II (ii)	Q.	7	8	9			
	A.	2	55	73			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	D	B	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C,D	A,C,D	A,B,D	A,C,D	A,B,C,D	C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.24	3.20	83.14	58.19 or 58.20	100.15	7.50
SECTION-II (ii)	Q.	7	8	9			
	A.	9	8	9			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	C	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	C,D	A,B,C,D	A,B,D	A,B,C,D	A,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	19.50	14.00	0.40	16.00	16.00	68.57
SECTION-II (ii)	Q.	7	8	9			
	A.	40	2	83			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (D)**

Given 10 VSD = 9 MSD

Here MSD → Main Scale division

$$1 \text{ VSD} = \frac{9}{10} \text{ MSD}$$

VSD → Vernier Scale division

Least count = 1 MSD – 1 VSD

$$= \left(1 - \frac{9}{10}\right) \text{ MSD}$$

$$= 0.1 \text{ MSD}$$

$$= 0.1 \times 0.1 \text{ cm}$$

$$= 0.01 \text{ cm}$$

As '0' of V.S. lie before '0' of M.S.

$$\text{Zero error} = -[10 - 6] \text{ L.C.}$$

$$= -4 \times 0.01 \text{ cm}$$

$$= -0.04 \text{ cm}$$

$$\text{Reading} = 4.1 \text{ cm} + 2 \times \text{LC}$$

$$= 4.1 \text{ cm} + 2 \times 0.01 \text{ cm}$$

$$= 4.12 \text{ cm}$$

$$\text{True diameter} = \text{Reading} - \text{Zero error}$$

$$= 4.12 - (-0.04) \text{ cm} = 4.16 \text{ cm}$$

2. Ans (B)

Process 1

$P = \text{constant}$, Volume increases and temperature also increases

$\Rightarrow W = \text{positive}$, $\Delta U = \text{positive}$

$\Rightarrow$ Heat is positive and supplied to gas

Process -2

$V = \text{constant}$, Pressure decrease

$\Rightarrow$ Temperature decreases

$$W = \int p dV = 0$$

ΔT is negative and $\Delta U = \frac{f}{2} nR\Delta T$

$\Rightarrow \Delta U$ is negative

$$\Delta Q = \Delta U + W$$

$\therefore \Delta Q \rightarrow$ Heat is negative and rejected by gas

Process 3

$P = \text{constant}$, Volume decreases

$\Rightarrow$ Temperature also decreases

$W = P\Delta V = \text{negative}$

$$\Delta U = \frac{f}{2} nR\Delta T = \text{negative}$$

$$\Delta Q = W + \Delta U = \text{negative}$$

Heat is negative and rejected by gas.

Process 4

$V = \text{constant}$, Pressure increases

$$W = \int p dV = 0$$

$PV = nRT \Rightarrow$ Temperature increase

$$\Rightarrow \Delta U = \frac{f}{2} nR\Delta T \text{ is positive}$$

$$\Delta Q = \Delta U + W$$

= positive

3. Ans (D)

Focal length of convex lens (f_1)

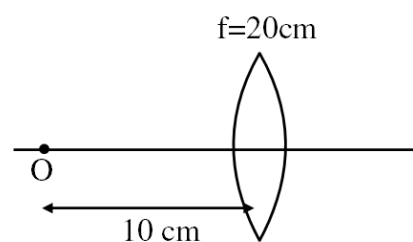
$$\begin{aligned} \frac{1}{f_1} &= (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] \\ &= (1.5 - 1) \left[\frac{1}{20} - \left(\frac{1}{-20} \right) \right] \\ \frac{1}{f_1} &= \frac{1}{20} \end{aligned}$$

$$\Rightarrow \boxed{f_1 = +20 \text{ cm}}$$

Focal length of concave lens (f_2)

$$\begin{aligned} \frac{1}{f_2} &= (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] \\ \frac{1}{f_2} &= (1.5 - 1) \left[-\frac{1}{20} - \frac{1}{20} \right] = \frac{1}{-20} \\ \Rightarrow \boxed{f_2 = -20 \text{ cm}} \end{aligned}$$

For lens 1



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \boxed{v = -20 \text{ cm}}$$

$$m_1 = \frac{v}{u} = \frac{-20}{-10} = 2$$

For lens 2

$$u = -40, f = -20, \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$v = -\frac{40}{3} \text{ cm}$$

$$m_2 = \frac{1}{3}$$

Net magnification

$$m = m_1 m_2 = \frac{2}{3}$$

4. Ans (C)

Out of 1000 nuclei of Q 40% may go α -decay

$\Rightarrow$ 400 nuclei may have α -decay

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{20}$$

$t = 1 \text{ hour} = 60 \text{ minutes}$

Using

$$N = N_0 e^{-\lambda t}$$

$$N = \frac{400}{8}$$

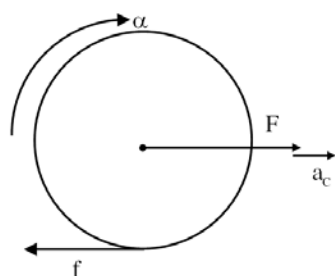
So, No. of nuclei decayed

$$= 400 - \frac{400}{8} = 350$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,D)



$$F - f = ma_c$$

$$fR = I_C \alpha$$

$$a_c - \alpha R = 0$$

$$F - I_C \frac{\alpha}{R} = ma_c$$

$$a_c = \frac{F}{\frac{I_C}{R^2} + m}$$

$$f = \frac{I_C \alpha}{R} = \frac{I_C}{R^2} a_c = \frac{I_C}{R^2} \frac{F}{\left[\frac{I_C}{R^2} + m \right]}$$

$$f = \frac{F}{\left(m + \frac{I_C}{R^2} \right)}$$

Thin walled hollow cylinder

$$I_C = mR^2$$

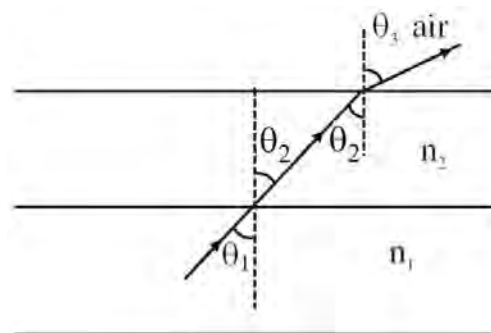
$$a_c = \frac{F}{2m}$$

$$fR = I_C \alpha = \frac{I_C a_c}{R}$$

$$f = \frac{I_C a_c}{R^2} \leq \mu mg$$

$$a_c \leq \frac{\mu mg R^2}{I_C}$$

6. Ans (A,B,C)



$$\sin \theta > \frac{1}{n_1} \text{ (Given)}$$

$$\text{i.e. } \sin \theta_1 > \frac{1}{n_1}$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\sin \theta_2 = \frac{n_1 \sin \theta_1}{n_2}$$

$$\text{If } n_1 = n_2 \text{ then } \theta_2 = \theta_1$$

$$n_2 \sin \theta_2 = (1) \sin \theta_3$$

$$\sin \theta_3 = n_2 \sin \theta_2$$

$$\sin \theta_3 = n_1 \sin \theta_1$$

$$\sin \theta_1 = \frac{\sin \theta_3}{n_1} > \frac{1}{n_1}$$

$$\sin \theta_3 > 1$$

$$\theta_3 > 90^\circ$$

This means ray cannot enter air

$$\text{For } n_1 > n_2; \sin \theta_1 = \frac{n_2}{n_1} \sin \theta_2 > \frac{1}{n_1}$$

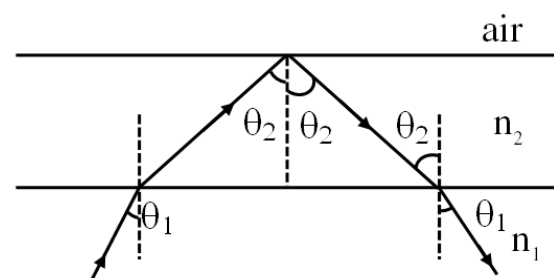
$$\sin \theta_2 > \frac{1}{n_2}$$

$$\text{for surface 2 - air interface } n_2 \sin \theta_2 = \sin \theta_3$$

$$\sin \theta_2 = \frac{\sin \theta_3}{n_2} > \frac{1}{n_2}$$

$$\theta_2 > 90^\circ$$

It means ray is reflected back in medium-2



for surface 1 – surface 2 interface

$$n_2 \sin \theta_2 = n_1 \sin \theta_1$$

$$\sin \theta_{2c} = \frac{n_1}{n_2}$$

θ_{2c} : critical angle

for ray to enter medium-1

$$\theta_2 < \theta_{2c}$$

$$\sin \theta_2 < \sin \theta_{2c}$$

$$\frac{n_1}{n_2} \sin \theta_1 < \frac{n_1}{n_2}$$

$$\sin \theta_1 < 1$$

$$\theta_1 < 90^\circ, \text{ which is true}$$

Hence ray enters medium-1

For $n_2 > n_1$

$$\frac{n_2}{n_1} \sin \theta_2 > \frac{n_2}{n_1}$$

$$\sin \theta_2 > \frac{1}{n_2}$$

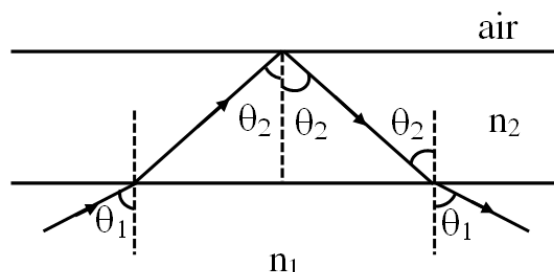
For surface 2 – air interface

$$n_2 \sin \theta_2 = \sin \theta_3$$

$$\sin \theta_2 = \frac{\sin \theta_3}{n_2} > \frac{1}{n_2}$$

$$\theta_2 > 90^\circ$$

It means ray is reflected back in medium - 2



$$n_2 \sin \theta_2 = n_1 \sin \theta_1$$

$$\sin \theta_1 = \frac{n_2}{n_1} \sin \theta_2$$

$$\sin \theta_{2c} = \frac{n_1}{n_2}; \theta_{2c} \rightarrow \text{critical angle}$$

For ray to enter medium - 1

$$\theta_2 < \theta_{2c}$$

$$\sin \theta_2 < \sin \theta_{2c}$$

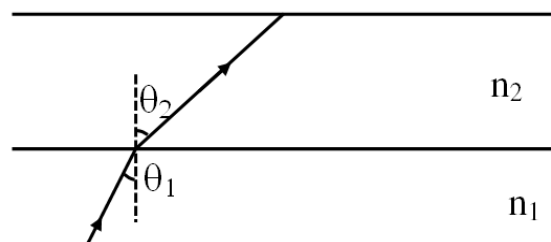
$$\frac{n_1}{n_2} \sin \theta_1 < \frac{n_1}{n_2}$$

$$\sin \theta_1 < 1$$

$$\theta_1 < 90^\circ, \text{ which is true}$$

Hence ray enters medium - 1

Let $n_2 = 1$



$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_2 = 1$$

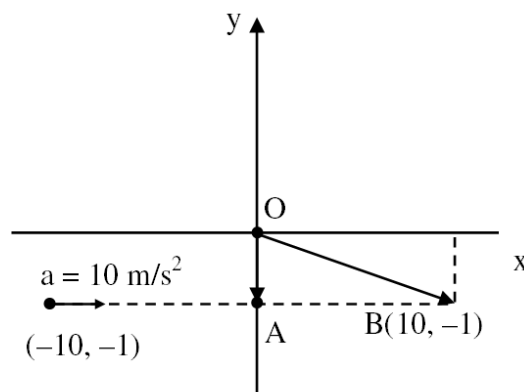
$$n_1 \sin \theta_1 = \sin \theta_2$$

$$\sin \theta_1 = \frac{\sin \theta_2}{n_1} > \frac{1}{n_1}$$

$$\sin \theta_2 > 1 \Rightarrow \theta_2 > 90^\circ$$

ray is reflected back in medium -

7. Ans (A,C,D)



$$\vec{r}_A = -\hat{j}$$

$$S = \frac{1}{2}at^2$$

$$10 = \frac{1}{2} \times 10 \times t^2$$

$$t = \sqrt{2} \text{ sec}$$

$$\vec{\tau}_0 = \vec{r} \times \vec{F}; \vec{r}_B = 10\hat{i} - \hat{j}$$

$$\vec{F} = m\vec{a} = 0.2 \times 10\hat{i} = 2\hat{i}$$

$$\vec{\tau}_0 = (10\hat{i} - \hat{j}) \times (2\hat{i})$$

$$\vec{\tau}_0 = 2\hat{k}$$

$$\vec{L}_0 = \vec{r}_B \times \vec{p} = \vec{r}_B \times m\vec{v}$$

$$\vec{v} = \vec{a}t = 10\hat{i} \times 2 = 20\hat{i}$$

$$\vec{L}_0 = (0.2) [(10\hat{i} - \hat{j}) \times 20\hat{i}] = 4\hat{k}$$

At point A(0, -1)

$$\vec{\tau}_0 = \vec{r}_A \times \vec{F} = (-\hat{j}) \times 2\hat{i} = 2\hat{k}$$

8. **Ans (A,B,C,D)**

For A

When the transition is from any level to $n = 2$, then photon emitted belong to Balmer series.

∴ For longest wavelength, transition occurs from $n = 3$ to $n = 2$.

$$\therefore \frac{hc}{\lambda_{\max}} = RCh \left[\frac{1}{2^2} - \frac{1}{3^2} \right] \quad \& \quad \text{for shortest}$$

wavelength transition occurs from $n = 3$ to $n = 2$

$$\therefore \frac{hc}{\lambda_{\min}} = RCh \left[\frac{1}{2^2} - \frac{1}{\infty^2} \right]$$

$$\therefore \frac{\lambda_{\text{longest}}}{\lambda_{\text{shortest}}} = \frac{9}{5}$$

For (B)

$$\lambda_{\text{longest of Balmer}} = \frac{36}{5R}$$

$$\lambda_{\text{longest of Paschen}} = \frac{9}{R}$$

Hence these wavelength don't overlap.

For (C)

For Lyman series,

$$\frac{1}{\lambda} = R \left[\frac{1}{1} - \frac{1}{m^2} \right]$$

$$\text{Also } \frac{1}{\lambda_0} = R$$

$$\therefore \frac{1}{\lambda} = \frac{1}{\lambda_0} \left[1 - \frac{1}{m^2} \right] \Rightarrow \lambda = \frac{\lambda_0}{1 - \frac{1}{m^2}}$$

For (D)

$$\lambda_{\text{longest of Lyman}} = \frac{4}{3R},$$

$$\lambda_{\text{shortest of Balmer}} = \frac{4}{R}$$

Hence that wavelength don't overlap.

9. **Ans (B,D)**

EMF developed across the emf of semi-circular

$$\text{rod} = \int_1^4 \frac{\mu_0 i}{2\pi r} dr = \frac{\mu_0 i v}{2\pi} \ln 4 = \frac{\mu_0 i v}{\pi} \ln 2$$

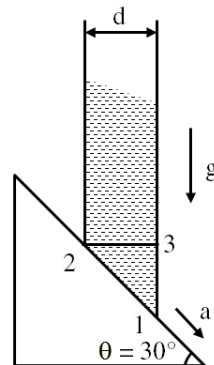
Form given value,

$$E = \frac{4\pi \times 10^{-7} \times 1 \times 3 \times 0.7}{\pi} = 12 \times 7 \times 10^{-8}$$

$$i_{\max} = \frac{E}{R} = \frac{12 \times 7 \times 10^{-8}}{1.4} = 0.6 \times 10^{-6} \text{ A}$$

$$Q_{\max} = C_0 E = 12 \times 7 \times 10^{-8} \times 5 \times 10^{-6} \\ = 4.2 \times 10^{-12} \text{ C}$$

10. **Ans (D)**



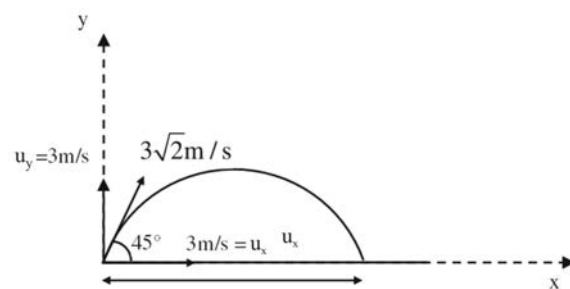
$$\therefore P_1 - P_3 = \rho \left(g - \frac{a}{2} \right) \frac{d}{\sqrt{3}}$$

$$P_2 - P_3 = \rho \frac{\sqrt{3}a}{2} d$$

PART-1 : PHYSICS

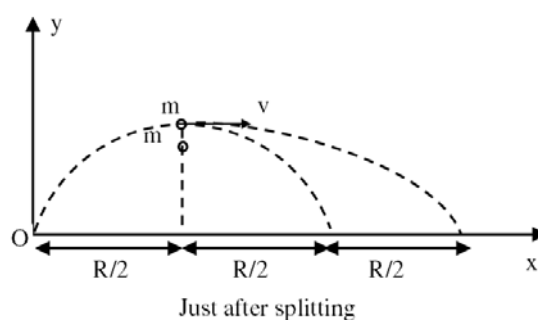
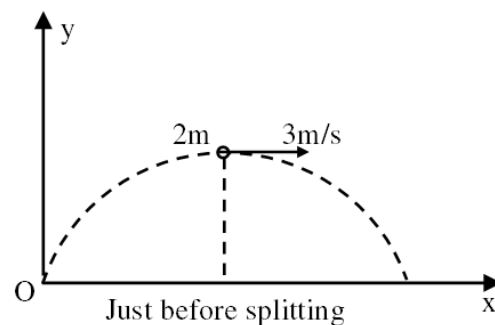
SECTION-II (i)

1. **Ans (0.30)**



$$\text{Range } R = \frac{2u_x u_y}{g} = \frac{2 \times 3 \times 3}{10} = 1.8 \text{ m}$$

$$\text{Time of flight } T = \frac{2u_y}{g} = \frac{2 \times 3}{10} = 0.6 \text{ sec}$$



∴ Time of motion of one part falling vertically downwards is $= 0.3 \text{ sec} = \frac{T}{2}$

⇒ Time of motion of another part, $t = \frac{T}{2} = 0.3 \text{ sec}$

From momentum conservation $\Rightarrow P_i = P_f$

$$2m \times 3 = m \times v$$

$$v = 6 \text{ m/s}$$

Displacement of other part in 0.3 sec in horizontal direction $= v \frac{T}{2}$

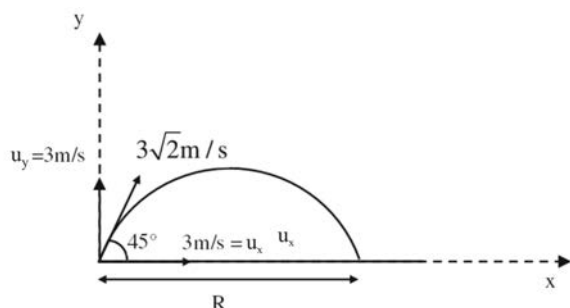
$$= 3 \times 0.3 = 0.9 \text{ m} = R$$

∴ Total distance of second part from point 'O' is,

$$x = \frac{3R}{2} = 3 \times \frac{1.8}{2}$$

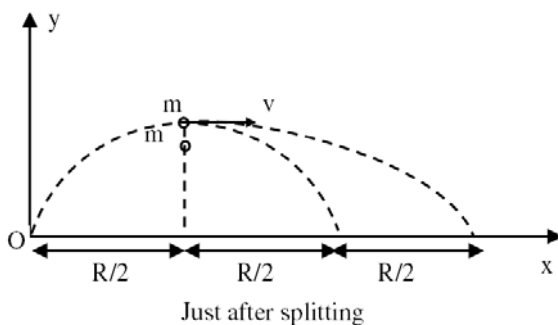
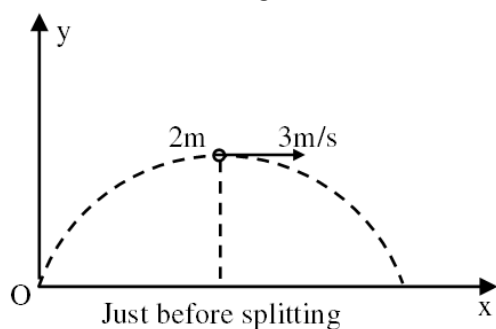
$$x = 2.7 \text{ m}$$

2. **Ans (2.70)**



$$\text{Range } R = \frac{2u_x u_y}{g} = \frac{2 \times 3 \times 3}{10} = 1.8 \text{ m}$$

$$\text{Time of flight } T = \frac{2u_y}{g} = \frac{2 \times 3}{10} = 0.6 \text{ sec}$$



∴ Time of motion of one part falling vertically downwards is $= 0.3 \text{ sec} = \frac{T}{2}$

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$$2m \times 3 = m \times v$$

$$v = 6 \text{ m/s}$$

Displacement of other part in 0.3 sec in horizontal direction $= v \frac{T}{2}$

$$= 3 \times 0.3 = 0.9 \text{ m} = R$$

∴ Total distance of second part from point 'O' is,

$$x = \frac{3R}{2} = 3 \times \frac{1.8}{2}$$

$$x = 2.7 \text{ m}$$

3. **Ans (2.00)**

When S is connected to position P

$$E_{eq} = \frac{\frac{4}{2} + \frac{1}{1}}{\frac{1}{1} + \frac{1}{2}} = 2$$

$$q_1 = C E_{eq} = 2$$

When S is connected to position Q

$$i = \frac{4}{3}$$

$$\text{P.d. across capacitor} = \frac{4}{3}$$

$$q_2 = \frac{4}{3}$$

4. **Ans (1.33)**

When S is connected to position P

$$E_{eq} = \frac{\frac{4}{2} + \frac{1}{1}}{\frac{1}{1} + \frac{1}{2}} = 2$$

$$q_1 = C E_{eq} = 2$$

When S is connected to position Q

$$i = \frac{4}{3}$$

$$\text{P.d. across capacitor} = \frac{4}{3}$$

$$q_2 = \frac{4}{3}$$

5. Ans (1.33)

$$\frac{KQ}{\sqrt{x^2 + y^2}} = \frac{KQ}{2\sqrt{(x-2)^2 + y^2}}$$

$$4(x-2)^2 + 4y^2 = x^2 + y^2$$

$$4x^2 + 16 - 16x + 4y^2 = x^2 + y^2$$

$$x^2 + y^2 - \frac{16x}{3} + \frac{16}{3} = 0$$

$$\left(x - \frac{8}{3}\right)^2 + y^2 = \frac{-16}{3} + \frac{64}{9} = \frac{16}{9}$$

6. Ans (2.66 to 2.67)

$$\frac{KQ}{\sqrt{x^2 + y^2}} = \frac{KQ}{2\sqrt{(x-2)^2 + y^2}}$$

$$4(x-2)^2 + 4y^2 = x^2 + y^2$$

$$4x^2 + 16 - 16x + 4y^2 = x^2 + y^2$$

$$x^2 + y^2 - \frac{16x}{3} + \frac{16}{3} = 0$$

$$\left(x - \frac{8}{3}\right)^2 + y^2 = \frac{-16}{3} + \frac{64}{9} = \frac{16}{9}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (2)

$$r = \frac{mv}{qB} = \frac{\sqrt{2mqV}}{qB}$$

$$\frac{p^2}{2m} = K.E = qV$$

$$\frac{r_p}{r_\alpha} \propto \sqrt{\frac{m}{q}}$$

8. Ans (55)

$$L = \frac{Ma^2}{3}\Omega + M\left(\frac{3a}{4}\right)^2\Omega + \frac{M\left(\frac{a}{4}\right)^2 8\Omega}{2}$$

$$L = \frac{55}{48}Ma^2\Omega$$

$$n = 55$$

9. Ans (73)

$$\sigma AT^4 = -ms \frac{dT}{dt}$$

$$\int_{400}^{200} \frac{dT}{T^4} = \int_0^{t_1} k dt$$

$$\frac{1}{3T^3} \Big|_{400}^{200} = kt_1$$

$$\frac{1}{3} \left(\frac{1}{200^3} - \frac{1}{400^3} \right) = kt_1$$

$$\frac{1}{3T^3} \Big|_{400}^{50} = kt_2$$

$$\frac{1}{3} \left(\frac{1}{50^3} - \frac{1}{400^3} \right) = kt_2$$

$$\frac{t_2}{t_1} = 73$$

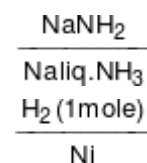
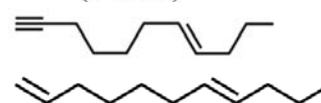
PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A)

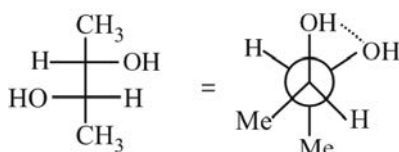


(1 mole)



Alkyne is more reactive than alkene towards hydrogenation.

2. Ans (D)



Most stable conformation with H-bonding

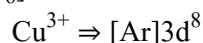
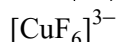
(2S, 3S) butane-2,3,-diol

(D) is the answer

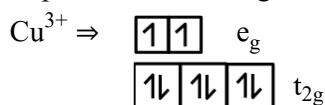
3. Ans (B)

$$\begin{aligned} d &= \frac{3 \times M_x + 1 \times M_y}{N_A \times a^3} \\ &= \frac{3 \times 10 + 1 \times 30}{6 \times 10^{23} \times 8 \times 10^{-24}} \\ &= 1.56 \end{aligned}$$

4. Ans (C)



In presence of F^- Ligand

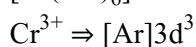


Number of unpaired electrons = 2

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

$$\mu = \sqrt{2(2+2)} \Rightarrow \sqrt{8} \text{ B.M.}$$

$$\Rightarrow 2.84 \text{ B.M.}$$



Number of unpaired electrons = 3

$$\mu = \sqrt{n(n+2)} \text{ B.M.}$$

$$\mu = \sqrt{3(3+2)} \text{ B.M.}$$

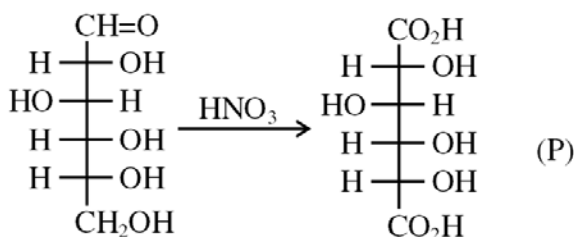
$$\mu = \sqrt{15} \text{ B.M.}$$

$$\Rightarrow 3.87 \text{ B.M.}$$

PART-2 : CHEMISTRY

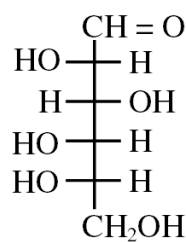
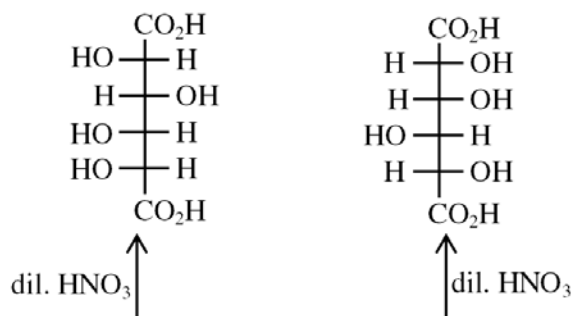
SECTION-I (ii)

5. Ans (B,C,D)

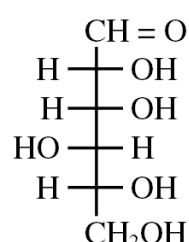


$$[\alpha]_D = 52.7^\circ$$

The enantiomer of P has rotation -52.7° is as follows

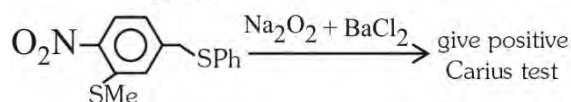
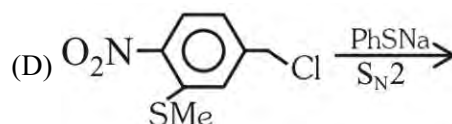
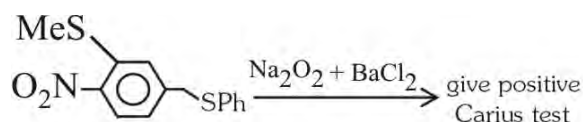
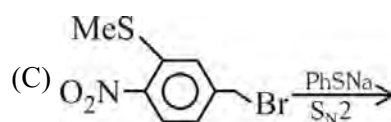
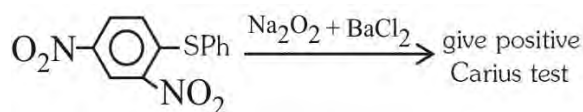
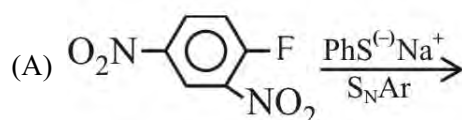


(C)



(D)

6. Ans (A,C,D)



7. Ans (A,B,D)

(A) Process of precipitating colloidal solution is called coagulation. Hence false.

(B) For colloidal solutions concentration is very small due to very large molar mass and hence their colligative properties are very small as compared to true solutions

$\therefore \Delta T_f$ is lesser for colloidal solution. Hence true.

(C) At CMC surfactant form micelles. Hence true

(D) Micelles and macromolecular colloids are two different types of colloids. Hence false.

8. **Ans (A,C,D)**

From state I to II (Reversible isothermal expansion)

⇒ P decreases, V increases, T constant

H constant & S increases.

From state II to III (Reversible adiabatic expansion)

⇒ P decreases, V increases, T decreases

H decreases, S constant

∴ Plots (A), (B), (D) are correct while (C) is wrong as from II to III, H is decreasing.

9. **Ans (A,B,C,D)**

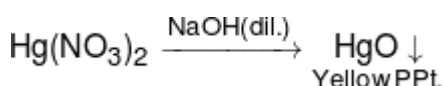
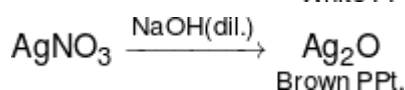
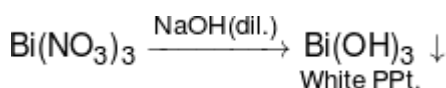
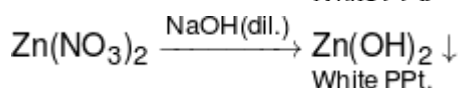
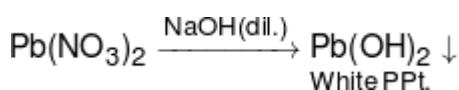
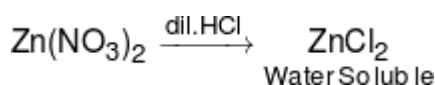
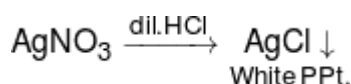
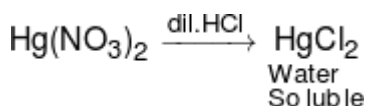
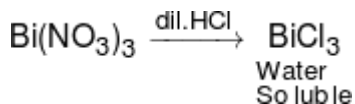
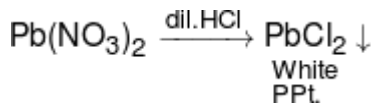
(A) $\text{PbS} + 2\text{PbO} \rightarrow 3\text{Pb} + \text{SO}_2$ (self reduction)

(B) Silica is added to remove impurity of Fe in the form of slag FeSiO_3

(C) CuFeS_2 ore is partially oxidized first by roasting and then self reduction of Cu takes place to produce blister copper.

(D) $4\text{Na}[\overset{+1}{\text{Ag}}(\text{CN})_2] + 2\text{Zn} \rightarrow 2\text{Na}_2[\overset{+2}{\text{Zn}}(\text{CN})_4] + 4\overset{0}{\text{Ag}}$
Reducing Agent

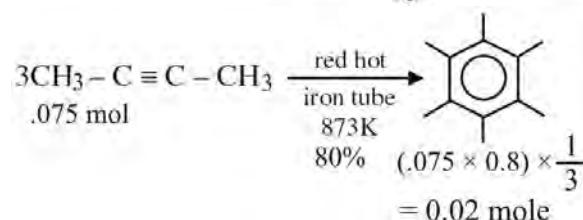
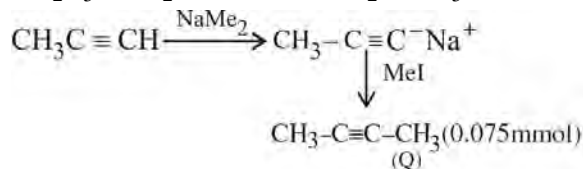
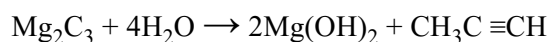
10. **Ans (C,D)**



PART-2 : CHEMISTRY

SECTION-II (i)

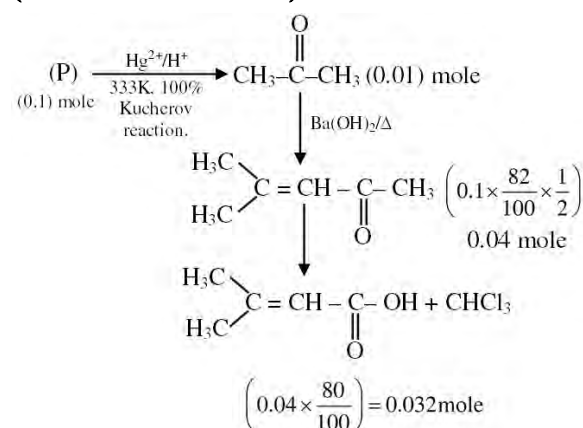
1. **Ans (3.24)**



The value of x = 162 × 0.02 = 3.24 gm

2. **Ans (3.20)**

(3.20 OR 3.90 TO 3.91)



$$60 + 32 + 8 = 100$$

The value of Y = 0.032 × 100 = 3.2

3. **Ans (83.14)**

$$\Delta G^\circ = -RT \ln \left(\frac{P}{1} \right) = \Delta H^\circ - T \Delta S^\circ$$

$$\ln \left(\frac{P}{1} \right) = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

$$\text{Slope} = -\frac{\Delta H^\circ}{R} = 10^4 \times \left(-\frac{5}{5} \right)$$

$$\Rightarrow \Delta H^\circ = 10^4 \times R$$

$$= 83.14 \text{ kJ/mole}$$

4. Ans (58.19 or 58.20)

From the plot when, $\frac{10^4}{T} = 10$

$$\Rightarrow T = 1000 \text{ K}$$

$$\ln\left(\frac{P_2}{P_1}\right) = -3$$

Substituting in equation :

$$\ln\left(\frac{P_2}{P_1}\right) = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

We get,

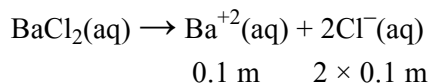
$$-3 = -\frac{10^4 \times R}{R \times 1000} + \frac{\Delta S^\circ}{R}$$

$$\Rightarrow \Delta S^\circ = 7R$$

$$\Rightarrow \Delta S^\circ = 7 \times 8.314 \text{ J/K-mol}$$

$$\Rightarrow \Delta S^\circ = 58.19 \text{ or } 58.20 \text{ J/K-mol}$$

5. Ans (100.15)

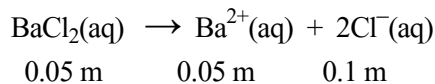
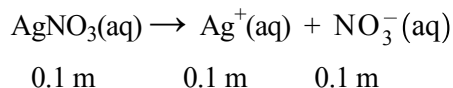


$$\Delta T_b = 0.3 \times 0.5$$

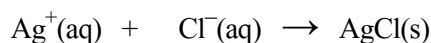
$$= 0.15^\circ\text{C} = 0.15 \text{ K}$$

$$\text{Boiling point of solution} = 100.15^\circ\text{C} = X$$

6. Ans (7.50)



Ag^+ and Cl^- combine to form AgCl precipitate



$$t = 0 \quad 0.1 \text{ m} \quad 0.1 \text{ m}$$

$$t = \infty \quad 0 \quad 0$$

In final solution total concentration of all ions :

$$[\text{NO}_3^-] + [\text{Ba}^{2+}] = 0.05 + 0.1 = 0.15 \text{ m}$$

$$\Delta T_b = 0.5 \times 0.15 = 0.075^\circ\text{C}$$

$$\text{B.P. of solution 'B'} = 100.075^\circ\text{C}$$

$$\text{B.P. of solution 'A'} = 100.15^\circ\text{C}$$

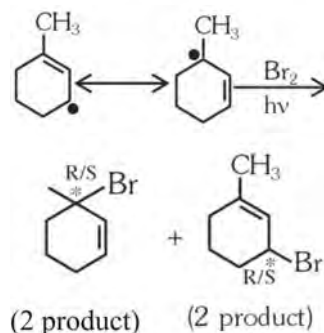
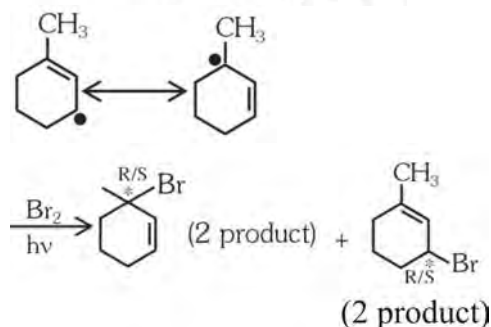
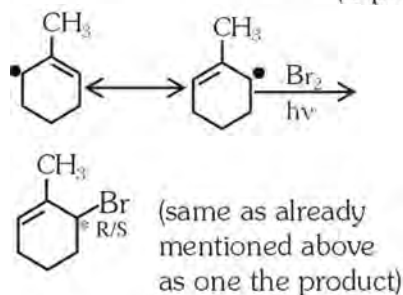
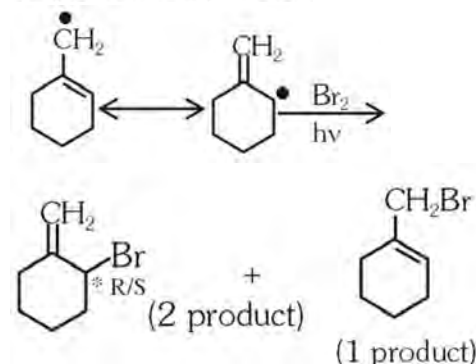
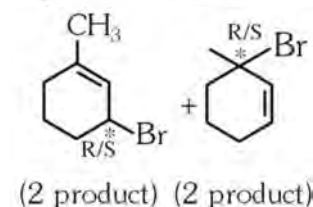
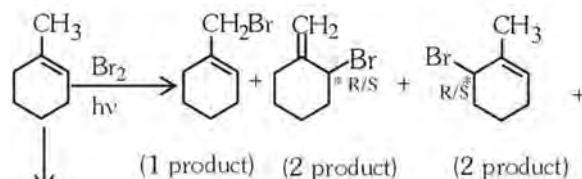
$$|\gamma| = 100.15 - 100.075$$

$$= 0.075 = 7.5 \times 10^{-2}$$

PART-2 : CHEMISTRY

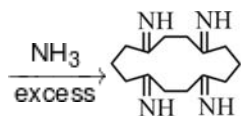
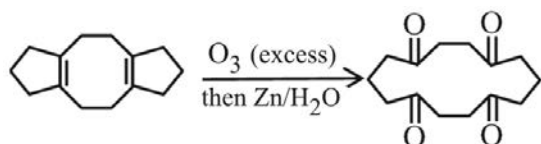
SECTION-II (ii)

7. Ans (9)



Total 9 product

8. Ans (8)



Total 8 atoms are sp^2 hybridised

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

Since (1, 1) lies on $y = x$ & it is orthocentre

$\Rightarrow D$ must be right angle at (1, 1)

so third side passing through (1, 1) $\perp$ to $y = x$

is $y + x - 2 = 0$

Diameter of circle : (-2, -2)

& $\left(\frac{5}{2}, \frac{-1}{2}\right)$: point of intersection of $x + y - 2 = 0$

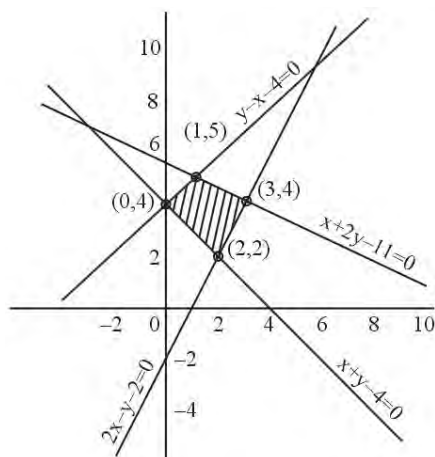
& $3y - x + 4 = 0$

Circle : $(x+2)^2 + (y+2)^2 = 0$

$\Rightarrow (x+2)^2 + (y+2)^2 = 0$

2. Ans (D)

For $R_1 : (y - x - 4)(y + x - 4) \leq 0$



$R_2 : (x + y - 4)(y - 2x + 2) \geq 0$

$R_3 : (2x - y - 2)(x + 2y - 11) \geq 0$

$$\text{Area} = \frac{1}{2} \begin{vmatrix} 0 & 2 & 3 & 1 & 0 \\ 4 & 2 & 4 & 5 & 4 \end{vmatrix}$$

$$= \frac{1}{2} (8 + 15 + 4 - 8 - 6 - 4)$$

$$= 4.5$$

3. Ans (C)

$$P = \frac{P(S_1 \cap (E_1 = E_3))}{P(E_1 = E_3)} = \frac{P(B_{1,3})}{P(B)}$$

$$P(B) = P(B_{1,3}) + P(B_{1,4}) + P(B_{3,4})$$

$\uparrow$ If 1,3 chosen at start
 $\uparrow$ If 1,4 chosen at start
 $\uparrow$ If 3,4 chosen at start

$$P(B_{1,3}) = \frac{1}{3} \times \frac{1 \times {}^3C_1}{{}^4C_2} \times \frac{1}{{}^5C_2}$$

1 is definitely chosen from F_2 1,3 chosen from G_2

$$P(B_{1,4}) = \frac{1}{3} \times \frac{1 \times {}^2C_1}{{}^3C_2} \times \frac{1}{{}^5C_2}$$

1 is definitely chosen from F_2 1,3 chosen from G_2

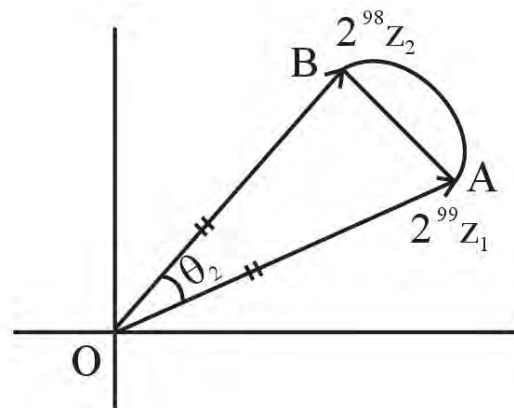
$$P(B_{3,4}) = \frac{1}{3} \times \left[\frac{{}^3C_2 \times 1}{{}^4C_2} \times \frac{1}{{}^4C_2} + \frac{1 \times {}^3C_1}{{}^4C_2} \times \frac{1}{{}^5C_2} \right]$$

If 1 is not chosen from F_2 If 1 is chosen from F_2

$$\frac{P(B_{1,3})}{P(B)} = \frac{1}{5}$$

4. Ans (A)

$$P : |z_2 - z_1| + |z_3 - z_2| + \dots + |z_{100} - z_{99}| + |z_1 - z_{100}|$$



$$\geq \|z_2\| - \|z_1\| + \|z_3\| - \|z_2\| + \dots + \|z_{100}\| - \|z_{99}\| +$$

$$\|z_{100}\| - \|z_1\|$$

$$= (2 - 1) + (2^2 - 2^1) + (2^3 - 2^2) + \dots + (2^{100} - 2^{99}) + (2^{100} - 1)$$

$$= 2^{100} - 1 + 2^{100} - 1 = 2^{101} - 2$$

$$\text{Arc} \geq |2^{98} z_2 - 2^{99} z_1|$$

$$\theta_2 \times OA \geq |2^{98} z_2 - 2^{99} z_1|$$

$$\Sigma \theta_2 \times 2^{99} \geq \Sigma |2^{98} z_2 - 2^{99} z_1|$$

$$\Rightarrow 2\pi \times 2^{99} \geq \Sigma |2^{98} z_2 - 2^{99} z_1|$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B,D)**

C is permutation matrix

$$CAC = B \text{ \& } C^3 = I, |A| = 0$$

$$|AD + C^2BD^{-1}| = |AD + C^2CACD^{-1}| = |AD + ACD^{-1}|$$

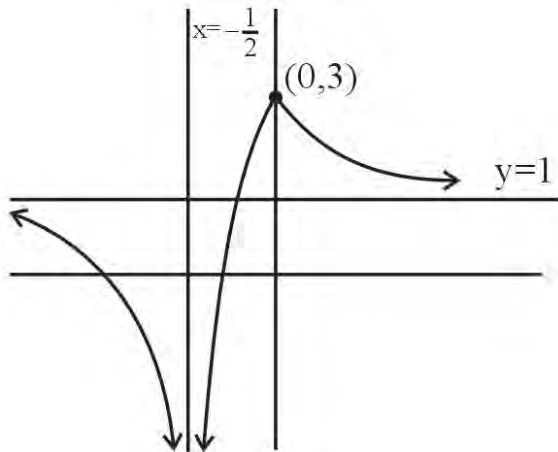
$$= A|D + CD^{-1}| = 0$$

$$|AD| + |CBD^{-1}| = |AD| + |C| \times |CAC| |D^{-1}|$$

$$= 0$$

6. **Ans (C,D)**

$$f(x) = \frac{4((2x+3)(4x^2+4x+1) - (2x+1)(4x^2+12x+3))}{(4x^2+4x+1)^2}$$



$$= \frac{4(-8x^2 - 4x)}{(4x^2 + 4x + 1)^2} = \frac{-16x(2x+1)}{(2x+1)^2}$$

$$= \frac{-16x}{2x+1}$$

$$\Rightarrow f(x) \geq 0$$

$$\left(-\infty, -\frac{1}{2}\right) \cup [0, \infty)$$

Asymptotes of graphs are $x = -\frac{1}{2}$ & $y = 1$

For $a > 0$ & two solution

$$\text{at } (0, 1) \cup (1, 3)$$

for b, for one solution : $b = 1, b = 3$

$$B = \{1, 3\}, |B| = 2$$

7. **Ans (A,B,C,D)**

$$P(E) = \frac{1}{6}; P(F) = \frac{1}{8}; P(G) = \frac{1}{4}; P(E \cap F \cap G) = \frac{1}{10}$$

$$(C) P(E \cup F \cup G) = P(E) + P(F) + P(G) - P(E \cap F) -$$

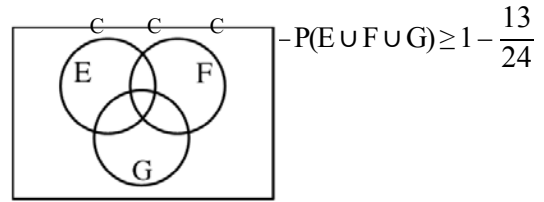
$$P(F \cap G) - P(G \cap E) + P(E \cap F \cap G)$$

$$= \frac{1}{6} + \frac{1}{8} + \frac{1}{4} - \sum P(E \cap F) + \frac{1}{10}$$

$$= \frac{3+4+6}{24} + \frac{1}{10} - \sum P(E \cap F)$$

$$= \frac{13}{24} + \frac{1}{10} - \sum P(E \cap F)$$

$$\Rightarrow P(E \cup F \cup G) \leq \frac{13}{24} \text{ [(C) is Correct]}$$



$$\Rightarrow P(E^C \cap F^C \cap G^C) \geq \frac{11}{24} \text{ [(D) is Incorrect]}$$

$$(A) P(E) = \frac{1}{6} \geq P(E \cap F \cap G^C) + P(E \cap F \cap G)$$

$$\Rightarrow \frac{1}{6} \geq P(E \cap F \cap G^C) + \frac{1}{10}$$

$$\Rightarrow \frac{1}{6} - \frac{1}{10} \geq P(E \cap F \cap G^C)$$

$$\Rightarrow \frac{1}{15} \geq P(E \cap F \cap G^C) \text{ [(A) is Correct]}$$

$$(B) P(F) = \frac{1}{8} \geq P(E^C \cap F \cap G) + P(E \cap F \cap G)$$

$$\Rightarrow \frac{1}{8} - \frac{1}{10} \geq P(E^C \cap F \cap G)$$

$$\Rightarrow \frac{1}{40} \geq P(E^C \cap F \cap G) \text{ [(B) is Correct]}$$

8. **Ans (A,B,D)**

$I + ABC$ and I commutes with D

$\Rightarrow (I + ABC) - I$ also commutes with D

$$\Rightarrow ABCD = DABC$$

$$\Rightarrow (ABCD)^T = C^T B^T A^T D^T$$

$$|I + ABC| |ADBC| = |(I + ABC) \times D| \times |ABC|$$

$$= 1 \times |ABC|$$

$$(I + CAB)(I - CDAB) = I + CAB - CDAB -$$

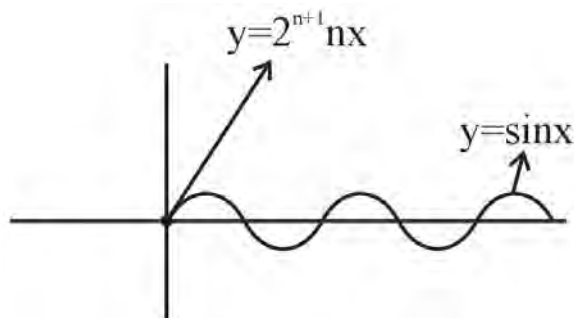
$$C(ABCD)AB$$

$$= I + CAB - CDAB - C(I - D)AB$$

$$= I$$

9. Ans (A,B,C,D)

$$f_n(n) = \sum_{r=1}^n \tan^{-1} \left(\frac{2^{r+1}rx - 2^r(r-1)x}{1 + (2^{r+1}rx)(2^r(r-1)x)} \right)$$



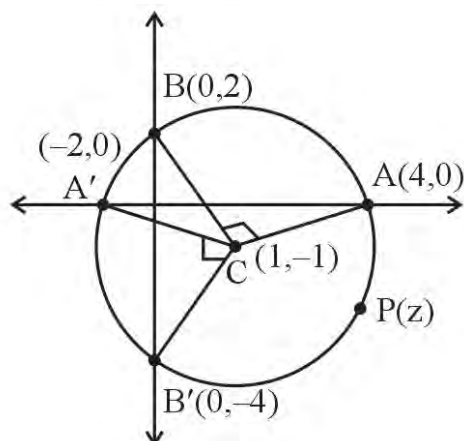
$$= \sum_{r=1}^n (\tan^{-1}(2^{r+1}rx) - \tan^{-1}(2^r(r-1)x))$$

$$= \tan^{-1}(2^{n+1}nx)$$

$$y = \tan(f_n(x)) = 2^{n+1}nx, \text{ slope} > 1 \text{ for } n \geq 1$$

10. Ans (A,D)

$$x = 0, y^2 + 2y - 8 = 0 \Rightarrow y = 2 \text{ \& } -4$$



$$y = 0, x^2 - 2x - 8 = 0 \Rightarrow x = -2 \text{ \& } 4$$

centre of circle (1, -1)

$$\angle A'CB' = \frac{\pi}{2}$$

$$\angle A'CB' = \frac{\pi}{2}$$

$$\text{For } \alpha = 4, \beta = 2$$

$$\text{For 'P' } \text{Arg} \left(\frac{z + 4i}{z + 2} \right) = \frac{\pi}{4}$$

$$\text{For } \alpha = -2, \beta = -4$$

$$\text{For P, } \text{Arg} \left(\frac{z - 2i}{z - 4} \right) = \frac{\pi}{4}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (19.50)

$P_1 = 1 - \text{prob}(\text{minimum of chosen numbers is at least } 61)$

$$= 1 - \left(\frac{40}{100} \right)^3 = 1 - \frac{8}{125} = \frac{117}{125}$$

2. Ans (14.00)

Let $A = \{1, 2, \dots, 40, \dots, 80\}$

$$B = \{1, 2, 3, \dots, 40\}$$

$$P_2 = P(A_3) - P(B_3) = \left(\frac{80}{100} \right)^3 - \left(\frac{40}{100} \right)^3 = \frac{56}{125}$$

3. Ans (0.40)

$$P_1 : x + y + z = a$$

$$P_2 : x + 5y - z = b$$

$$P_3 : x - y + 2z = c + 2$$

for consistent

$$P_2 = 3P_1 - 2P_3 \Rightarrow b = 3a - 2c - 4$$

$$\Rightarrow 3a - b - 2c - 4 = 0$$

$$\text{so } P : 3x - y - 2z - 4 = 0$$

$$D = \frac{4}{\sqrt{14}}$$

$$M = \begin{vmatrix} a & -2 & b \\ 2 & 1 & c \\ 1 & 0 & 3 \end{vmatrix}$$

$$= 3a - 2(c - 6) + b(-1)$$

$$= 3a - b - 2c + 12$$

$$= 4 + 12 = 16$$

4. Ans (16.00)

$$P_1 : x + y + z = a$$

$$P_2 : x + 5y - z = b$$

$$P_3 : x - y + 2z = c + 2$$

for consistent

$$P_3 = 3P_1 - 2P_2 \Rightarrow b = 3a - 2c - 4$$

$$\Rightarrow 3a - b - 2c - 4 = 0$$

$$\text{so } P : 3x - y - 2z - 4 = 0$$

$$D = \frac{4}{\sqrt{14}}$$

$$M = \begin{vmatrix} a & -2 & b \\ 2 & 1 & c \\ 1 & 0 & 3 \end{vmatrix}$$

$$= 3a - 2(c - 6) + b(-1)$$

$$= 3a - b - 2c + 12$$

$$= 4 + 12 = 16$$

5. Ans (16.00)

$$P(x, y) \left| \frac{\sqrt{2}x + y - 1}{\sqrt{3}} \right| = \lambda^2$$

$$\left| \frac{2x^2 - (y-1)^2}{3} \right| = \lambda^2, C : |2x^2 - (y-1)^2| = 3\lambda^2$$

$$\text{line } y = 2x + 1, RS = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2},$$

$$R(x_1, y_1) \text{ and } S(x_2, y_2)$$

$$y_1 = 2x_1 + 1 \text{ and } y_2 = 2x_2 + 1 \Rightarrow (y_1 - y_2) = 2(x_1 - x_2)$$

$$RS = \sqrt{5(x_1 - x_2)^2} = \sqrt{5} |x_1 - x_2|$$

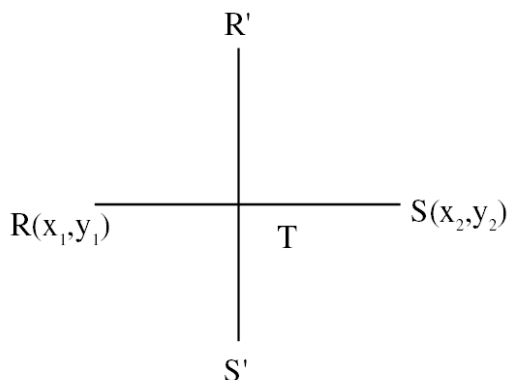
solve curve C and line $y = 2x + 1$ we get

$$|2x^2 - (2x)^2| = 3\lambda^2 \Rightarrow x^2 = \frac{3\lambda^2}{2}$$

$$RS = \sqrt{5} \left| \frac{2\sqrt{3}\lambda}{\sqrt{2}} \right| = \sqrt{30}\lambda = \sqrt{480} \Rightarrow 30\lambda^2 = 480 \Rightarrow \lambda^2 = 16$$

6. Ans (68.57)

$\perp$ bisector of RS



$$T \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{Here } x_1 + x_2 = 0$$

$$T = (0, 1)$$

Equation of

$$R'S' : (y - 1) = -\frac{1}{2}(x - 0) \Rightarrow x + 2y = 2$$

$$R'(a_1, b_1) S'(a_2, b_2)$$

$$D = (a_1 - a_2)^2 + (b_1 - b_2)^2 = 5(b_1 - b_2)^2$$

$$\text{solve } x + 2y = 2 \text{ and } |2x^2 - (y - 1)^2| = 3\lambda^2$$

$$|8(y - 1)^2 - (y - 1)^2| = 3\lambda^2 \Rightarrow (y - 1)^2 = \left(\frac{\sqrt{3}\lambda}{\sqrt{7}} \right)^2$$

$$y - 1 = \pm \frac{\sqrt{3}\lambda}{\sqrt{7}} \Rightarrow b_1 = 1 + \frac{\sqrt{3}\lambda}{\sqrt{7}}, b_2 = 1 - \frac{\sqrt{3}\lambda}{\sqrt{7}}$$

$$D = 5 \left(\frac{2\sqrt{3}\lambda}{\sqrt{7}} \right)^2 = \frac{5 \times 4 \times 3\lambda^2}{7} = \frac{5 \times 4 \times 3 \times 16}{7} = 68.57$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (40)

Case 1 : when

For '3' solution

$$y = x^2 - 4x + a \text{ touches}$$

$$y = 8 - 2x^2 \text{ in } [-2, 2]$$

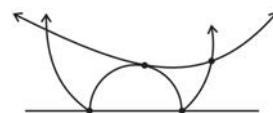
$$\Rightarrow x^2 - 4x + a = 8 - 2x^2$$

$$\Rightarrow 3x^2 - 4x + a - 8 = 0 \text{ has exactly one root}$$

$$16 - 12(a - 8) = 0$$

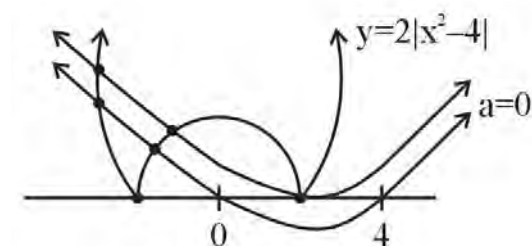
$$\Rightarrow 4 - 3a + 24 = 0 \Rightarrow 28 = 3a$$

$$\Rightarrow a = \frac{28}{3}$$



(Note : L.R. of $y = x^2 - 4x + a >$ L.R. of $y = 2x^2 - 8$)

Case- 2 :



when $y = x^2 - 4x + a$ passes through (2, 0)

$$a = 4$$

8. **Ans (2)**

$$\begin{aligned}
 &= \frac{(\cot A \cot C)}{\frac{(\cot A \cot C - 1)}{(\cot A + \cot C)}} = \frac{\cot A + \cot C}{-\cot(A + C)} \\
 &= \frac{\cot A + \cot C}{\cot B} = \frac{2b^2}{a^2 + c^2 - b^2} = \frac{2 \times 22}{21 + 23 - 22} \\
 &= \frac{2 \times 22}{22} = 2
 \end{aligned}$$

 9. **Ans (83)**

$$\begin{aligned}
 [\vec{a} \ \vec{b} \ \vec{c}]^2 &= \left(\frac{\sqrt{85}}{3} \right)^2 \\
 \Rightarrow \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} &= \frac{85}{9} \\
 \Rightarrow \begin{vmatrix} 1 & \vec{a} \cdot \vec{b} & 2 \\ \vec{a} \cdot \vec{b} & 2 & 1 \\ 2 & 1 & 9 \end{vmatrix} &= \frac{85}{9} \\
 \Rightarrow \vec{a} \cdot \vec{b} &= \frac{2}{9} \\
 |\vec{a} + \vec{b} + \vec{c}|^2 &= a^2 + b^2 + c^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \\
 &= 1 + 2 + 9 + 2\left(\frac{2}{9} + 1 + 2\right) \\
 &= 18 + \frac{4}{9}
 \end{aligned}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

19-03-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,D	A,B,D	B,C,D	A,B,D	A,B	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	A	C	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.90	3.91 to 3.92	0.45	0.25	20.00	1.04 to 1.05
SECTION-II (ii)	Q.	7	8	9			
	A.	6	3	7			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C,D	A,C,D	B,C,D	A,C	A,B	A,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	C	A	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.26 or 0.27	1.06 or 1.07	7.14	2.46	2.25	22.50
SECTION-II (ii)	Q.	7	8	9			
	A.	900	33	11			

PART-3 : MATHEMATICS

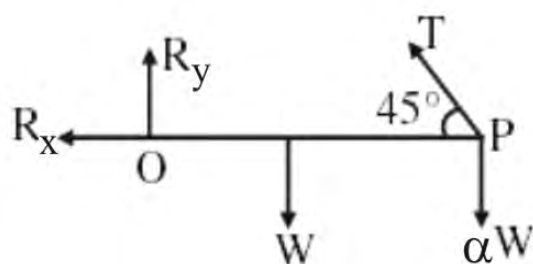
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,C	C	A,B,D	B,D	A,D	B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	D	D	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.50	3.00	50.00	4.00	3.00	16.50
SECTION-II (ii)	Q.	7	8	9			
	A.	2140	5	155			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B,D)



$$R_y + \frac{T}{\sqrt{2}} = W + \alpha W \quad \dots(i)$$

$$R_x = \frac{T}{\sqrt{2}} \quad \dots(ii)$$

Taking torque about 'O'

$$W \frac{l}{2} + \alpha W l = \frac{T}{\sqrt{2}} l$$

$$T = \sqrt{2} \left(\frac{W}{2} + \alpha W \right) \quad \dots(iii)$$

$$R_x = \frac{T}{\sqrt{2}} = \left(\frac{W}{2} + \alpha W \right)$$

Taking torque about P

$$R_y \ell = W \frac{\ell}{2}$$

$$R_y = \frac{W}{2}$$

when $T = T_{\max}$

$$4\sqrt{2}W = \sqrt{2} \left(\frac{W}{2} + \alpha W \right)$$

$$\text{we get } \alpha = \frac{7}{2}$$

2. **Ans (A,B,D)**

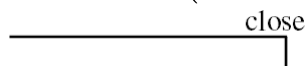
$$f = f_s \left(\frac{v}{v-u} \right)$$

$$(A) f = f_0 \left(\frac{v}{v-0.8v} \right) = 5f_0$$

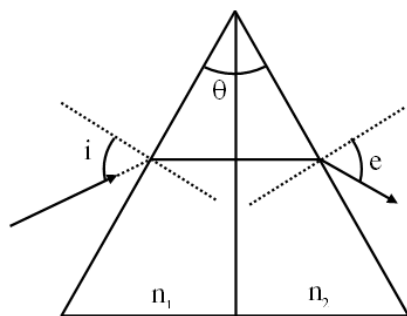
$$(B) f = 3f_0 \left(\frac{v}{v-0.8v} \right) = 15f_0$$

$$(C) f = 2f_0 \left(\frac{v}{v-0.5v} \right) = 4f_0$$

$$(D) f = 1.5f_0 \left(\frac{v}{v-0.5v} \right) = 3f_0$$



3. **Ans (B,C,D)**



$$\sin i = \frac{\sqrt{3}}{2}$$

$$i = 60^\circ$$

$$n_2 \sin 30^\circ = \sin e$$

$$\Delta n_2 \sin 30^\circ = \cos e \Delta e$$

$$\frac{\Delta n_2}{2} = \frac{1}{2} \Delta e$$

4. **Ans (A,B,D)**

$$\vec{S} = [\vec{E} \times \vec{B}] \frac{1}{\mu_0}$$

$\vec{S}$ is pointing vector denotes flow of energy per unit area per unit time

$$\vec{S} = \frac{\text{watt}}{\text{m}^2}$$

5. **Ans (A,B)**

$$N \rightarrow P + Q$$

$$\text{Energy released} = (m_N - m_P - m_Q)c^2 = \delta c^2$$

This will be distributed kinetic energy of P and Q

$$\Rightarrow E_P + E_Q = \delta c^2 \quad \dots(i)$$

By conservation of momentum

$$v_P = \frac{p}{m_P} \quad \leftarrow \text{P} \quad \text{Q} \quad \rightarrow \frac{p}{m_Q} = v_Q$$

$$\text{So, } \frac{v_P}{v_Q} = \frac{M_Q}{M_P} \quad \dots(ii)$$

$$\text{Kinetic energy be written as } KE = \frac{p^2}{2m}$$

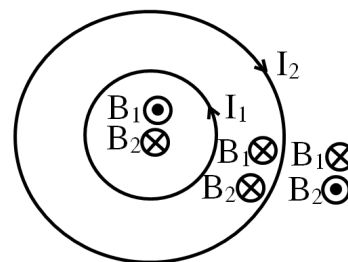
Hence divided in inverse ratio of masses.

$$E_P = \frac{M_Q}{M_P + M_Q} \delta c^2 \quad \dots(iii)$$

$$\text{By equation (i)} \Rightarrow \frac{p^2}{2M_P} + \frac{p^2}{2M_Q} = \delta c^2$$

$$\Rightarrow \frac{p^2}{2\mu} = \delta c^2 \Rightarrow p = c\sqrt{2\mu\delta}$$

6. **Ans (A,B,C)**



$$(A) d\vec{B} = \frac{\mu_0 I d\vec{\ell} \times \vec{r}}{4\pi r^3}$$

$\vec{d\ell}$ is in xy plane & $\vec{r}$ is also in xy plane

so $d\vec{B}$ is perpendicular to xy plane

(B) Due to symmetry it depends only on

$$r = \sqrt{x^2 + y^2}$$

$$(C) \text{ At centre } B_1 = \frac{\mu_0 I_1}{2R}; B_2 = \frac{\mu_0 I_2}{4R} \Rightarrow B_2 > B_1$$

but as we approach towards first loop B_1 increases to infinity hence B_1 dominates. So it would be zero at some point between inner loops and centre.

PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (A)

$$\phi = Li = \frac{\mu_0 m}{2\pi r^3} \times \pi a^2$$

$$\Rightarrow i = \frac{\mu_0 m \pi a^2}{2\pi r^3 L}$$

$$\Rightarrow i \propto \frac{m}{r^3}$$

$$m' = \pi a^2 i = \frac{\mu_0 m \pi^2 a^4}{2\pi r^3 L}$$

$$F = \frac{k m^2 \pi^2 a^4}{2\pi r^7 L}$$

$$W = \int F dr \propto \int \frac{m^2 dr}{r^7}$$

$$W \propto \frac{m^2}{r^6}$$

10. Ans (C)

$$\text{Finally } V_L = \frac{7V_0}{4}, V_R = \frac{V_0}{4}$$

$$C_V = \frac{R}{\gamma - 1} = 2R \Rightarrow \gamma - 1 = \frac{1}{2}$$

$$\gamma = \frac{3}{2}$$

$$T_0 V_0^{\gamma-1} = T_R \left(\frac{V_0}{4} \right)^{\gamma-1}$$

$$\frac{T_R}{T_0} = 2$$

$$\rho \left(\frac{V_0}{4} \right)^\gamma = P_0 V_0^\gamma \Rightarrow P = P_0 \times 8$$

$$\frac{P V}{T_L} = \frac{P_0 V_0}{T_0} \Rightarrow T_L = 14 T_0$$

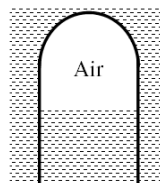
$$Q = n C_V \Delta T_1 + n C_V \Delta T_2$$

$$= 28 R T_0$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (0.90)



When it starts sinking

$$F_B = mg$$

$$\rho_0 (V_{\text{glass}} + V_{\text{gas}}) = m$$

$$1(2.3 + V_{\text{gas}}) = 4.6 \Rightarrow V_{\text{gas}} = 2.3 \text{ cc}$$

$$\text{Hence } \Delta V = 0.9 \text{ cc.}$$

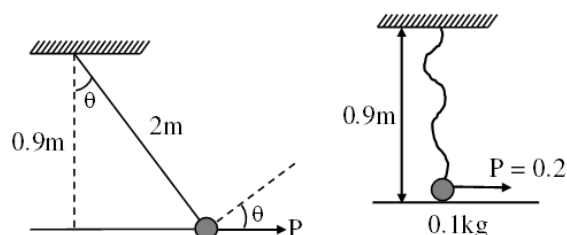
2. Ans (3.91 to 3.92)

Isothermal process for air

$$P_1 V_1 = P_2 V_2$$

$$\Delta P = 3.91 \times 10^4 \text{ Pa}$$

4. Ans (0.25)



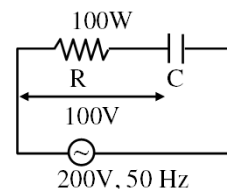
$$L = P \times 0.9 = 0.45 \text{ kgm}^2/\text{s}$$

Just after string becomes taut; there will be no velocity along the string.

$$\therefore V_1 = \frac{P \cos \theta}{m} = \frac{0.5 \times 0.9}{2 \times 0.1} = 2.25 \text{ m/s}$$

$$\therefore K = \frac{1}{2} m v_1^2 \approx 0.25$$

6. Ans (1.04 to 1.05)



$$i = 1$$

$$V_C = 100\sqrt{3}$$

$$X_C = 100\sqrt{3}$$

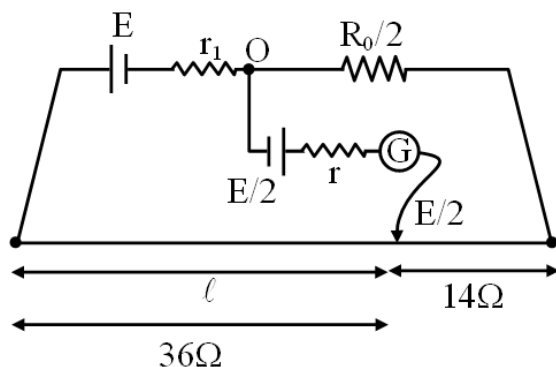
$$\frac{10^6}{2\pi 50 C} = 100\sqrt{3}$$

$$C = 20$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (6)



$$i \left(\frac{R_0}{2} + 0.28R_0 \right) = \frac{E_0}{2}$$

$$i \times 0.78R_0 = \frac{E_0}{2}$$

$$i = \frac{E_0}{2 \times 0.78R_0} = \frac{E_0}{r_1 + \frac{3}{2}R_0}$$

$$r_1 + 1.5 R_0 = 1.56 R_0$$

$$r_1 = 0.06 R_0$$

$$= 0.06 \times 100 = 6 \Omega$$

8. Ans (3)

Circular orbits

$$T = 2\pi \sqrt{\frac{R^3}{GM_S}}$$

Binary stars

$$nT = 2\pi \sqrt{\frac{(6R)^3}{G(8M_S + 16M_S)}}$$

$$n \times 2\pi \sqrt{\frac{R^3}{GM_S}} = 3 \times 2\pi \sqrt{\frac{R^3}{GM_S}}$$

$$n = 3$$

9. Ans (7)

$$E_1 - 4 = 2(E_1 - 5) = 2(E_2 - 6)$$

$$E_1 = 6$$

$$2 = (E_2 - 6)2$$

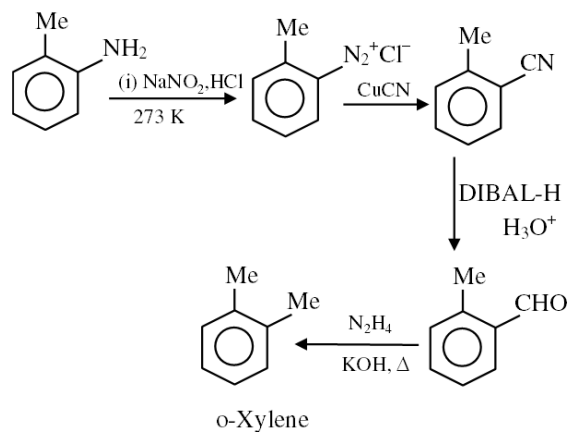
$$E_2 = 7$$

PART-2 : CHEMISTRY

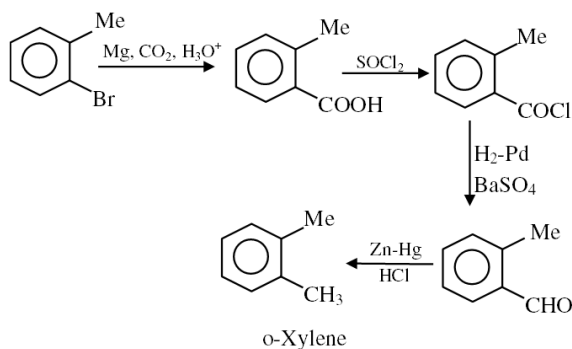
SECTION-I (i)

1. Ans (C,D)

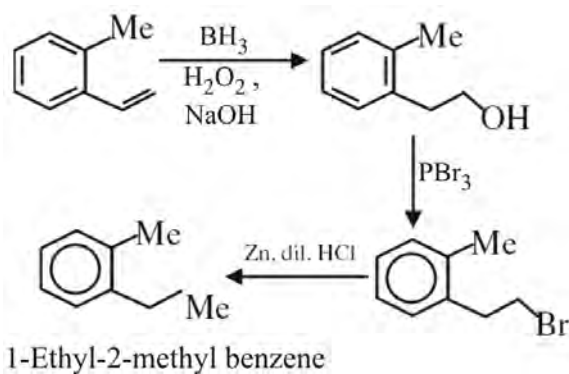
(A)



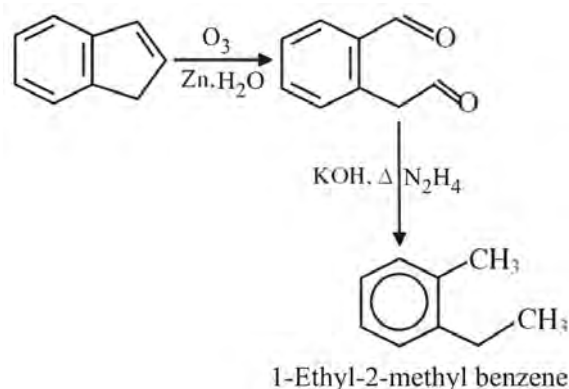
(B)



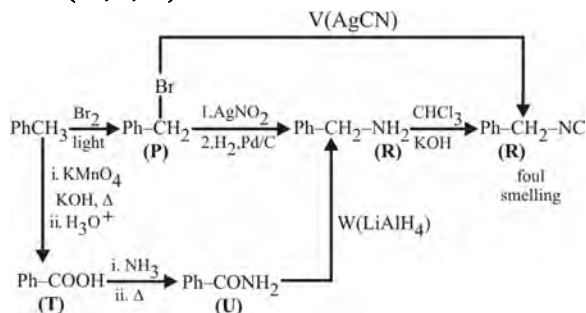
(C)



(D)



2. Ans (A,C,D)



3. Ans (B,C,D)

$$\frac{dp}{dt} = k[x]^1$$

$$3x + y \rightarrow p$$

$t = 0$ 3 1

$t = 50 \text{ s}$ $(3-1.5)$ $(1-0.5)$ 0.5

$$-\frac{1}{3} \frac{dx}{dt} = \frac{dp}{dt} = k[x]^1$$

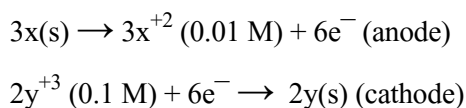
$$-\frac{dx}{dt} = 3k[x]^1$$

$$3k = \frac{\ln 2}{50} \Rightarrow k = \frac{\ln 2}{150}$$

At 50 sec $-\frac{dx}{dt} = 3k \times (1)^1 = \frac{\ln 2}{50}$

At 100 sec $-\frac{1}{3} \frac{dx}{dt} = \frac{-dy}{dt}$

4. Ans (A,C)



$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.06}{6} \log \frac{(x^{+2})^3}{(y^{+3})^2}$$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} + 0.06 \times \frac{2}{3}$$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} + 0.04$$

(A) Cd and Au $E^{\circ}_{\text{cell}} = +0.4 + 1.50$; $E_{\text{cell}} = 1.54$

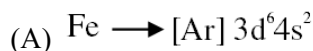
(B) Cd and Cr $E^{\circ}_{\text{cell}} = -0.34$; $E_{\text{cell}} = -0.30$

(C) Ni and Au $E^{\circ}_{\text{cell}} = 1.26$; $E_{\text{cell}} = 1.22$

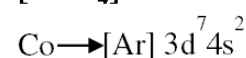
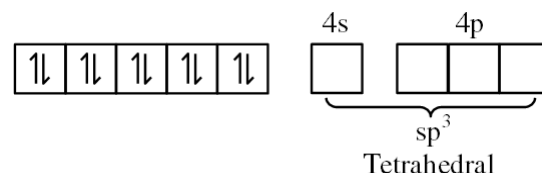
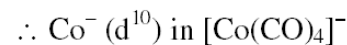
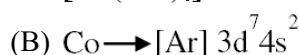
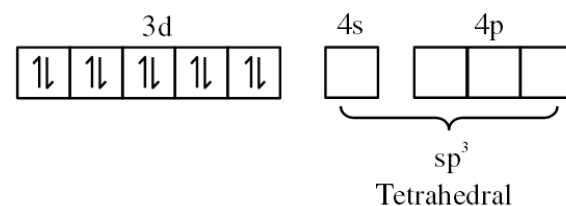
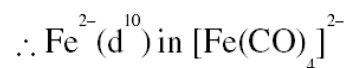
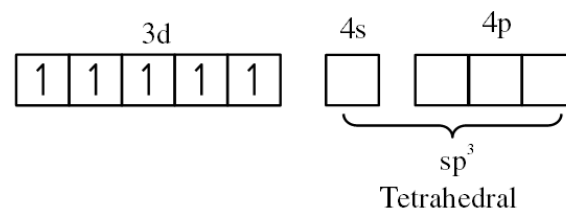
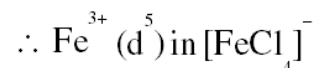
(D) Ni and Cr $E^{\circ}_{\text{cell}} = -0.5$; $E_{\text{cell}} = -0.46$

since in (A) & (C) E_{cell} is positive hence answer is (A) & (C).

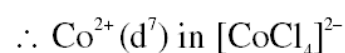
5. Ans (A,B)

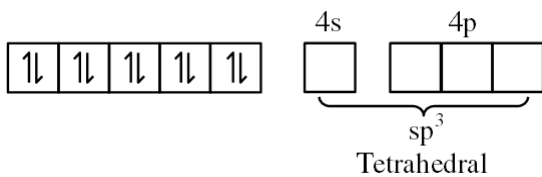


Cl^- is W.F.L. and does not pair up the unpaired electron of central metal atom.

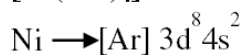
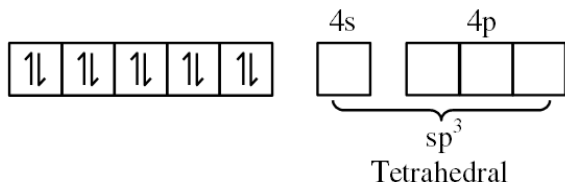
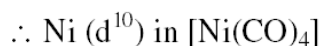
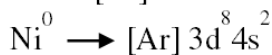
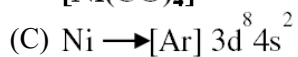
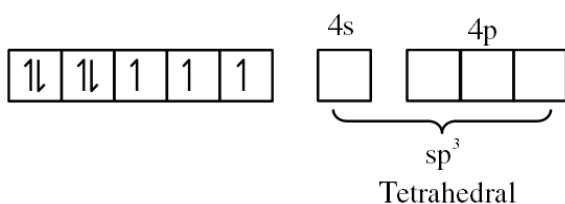
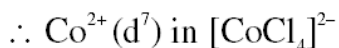


Cl^- is W.F.L. and does not pair up the unpaired electron of central metal atom.

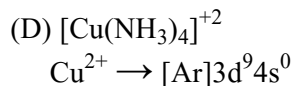
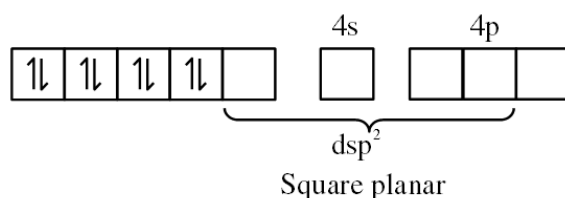
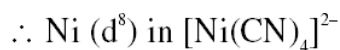




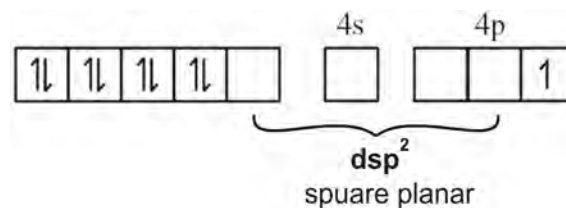
Cl^- is W.F.L. and does not pair up the unpaired electron of central metal atom.



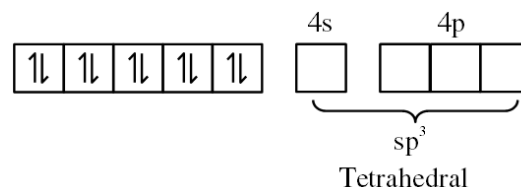
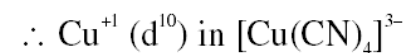
CN^- is S.F.L. and pair up the unpaired electron of central metal atom.



$\therefore NH_3$ is SFL



CN^- is S.F.L. and pair up the unpaired electron of central metal atom.

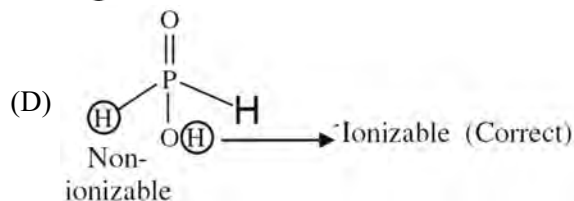
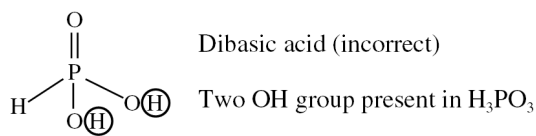


6. Ans (A,C,D)

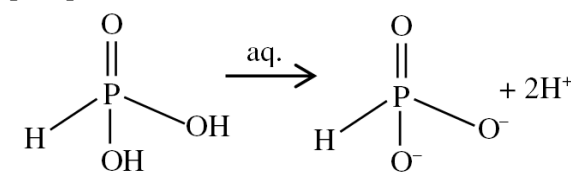


(B) H_3PO_4 has "P" in its highest oxidation state, hence cannot act as a reducing agent (correct)

(C)



The hydrogen which is directly attached to phosphorous does not ionized in water.

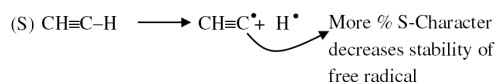
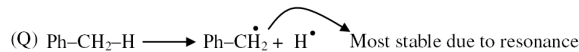
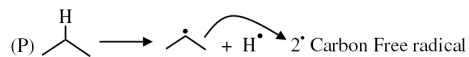


PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (C)**

Most stability of radical, less is the bond energy



Q require least BDE and S Required maximum BDE

Max BDE

So, Order of BDE $Q < P < R < S$

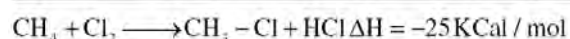
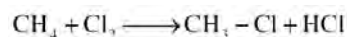
8. **Ans (C)**

Initiation step is endothermic hence option (A) is wrong.

Propagation step involving $\cdot\text{CH}_3$ formation is endothermic hence option (B) is wrong.

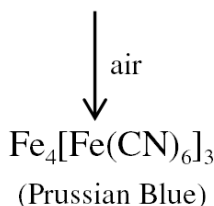
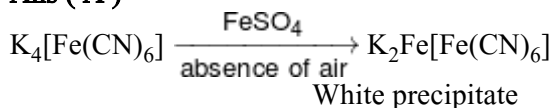
Propagation step involving CH_3Cl formation is exothermic hence option (C) is wrong.

Reaction

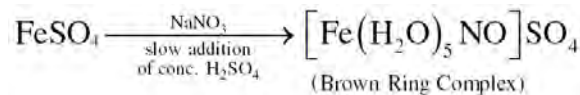


Overall reaction is exothermic with $\Delta H^\circ = -25$ KCal/mol, hence option (D) is correct.

9. **Ans (A)**



10. **Ans (D)**



PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (0.26 or 0.27)**

$$K_a = \frac{4\Lambda_m^3 C^2}{(\Lambda_m^0)^2 (\Lambda_m^0 - \Lambda_m)}$$

$$K_a = \frac{(y \times 10^2)^3 \times 4C^2}{(4 \times 10^2)^2 (4 \times 10^2 - y \times 10^2)}$$

$$= \frac{(3y \times 10^2)^3 \times \frac{4C^2}{100}}{(4 \times 10^2)^2 (4 \times 10^2 - 3y \times 10^2)}$$

$$\Rightarrow \frac{1}{(4-y)} = \frac{27}{100(4-3y)} \Rightarrow y = \frac{292}{273}$$

$$\alpha = \frac{\frac{44}{51} \times 10}{4 \times 10^2}$$

$$\alpha = 0.26 \text{ or } 0.27$$

$$y = 1.06 \text{ or } 1.07$$

2. **Ans (1.06 or 1.07)**

$$K_a = \frac{4\Lambda_m^3 C^2}{(\Lambda_m^0)^2 (\Lambda_m^0 - \Lambda_m)}$$

$$K_a = \frac{(y \times 10^2)^3 \times 4C^2}{(4 \times 10^2)^2 (4 \times 10^2 - y \times 10^2)}$$

$$= \frac{(3y \times 10^2)^3 \times \frac{4C^2}{100}}{(4 \times 10^2)^2 (4 \times 10^2 - 3y \times 10^2)}$$

$$\Rightarrow \frac{1}{(4-y)} = \frac{27}{100(4-3y)} \Rightarrow y = \frac{292}{273}$$

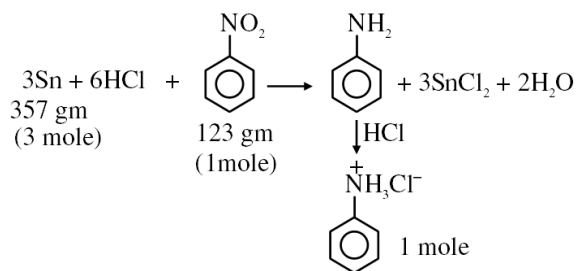
$$\alpha = \frac{\frac{44}{51} \times 10}{4 \times 10^2}$$

$$\alpha = 0.26 \text{ or } 0.27$$

$$y = 1.06 \text{ or } 1.07$$

3. **Ans (7.14)**

The value of x is



$$(72 + 8 + 35) + 14 = 129 \text{ gm (molecular weight of organic salt)}$$

So to get 2.58 gm organic salt.

We have to form 0.02 mole salt.

So 0.02 mole nitrobenzene is required.

0.06 mole Sn is required.

the amount of Sn required

$$= 0.02 \times 119 \times 3 = 7.14 \text{ gm}$$

So x is 7.14 gm

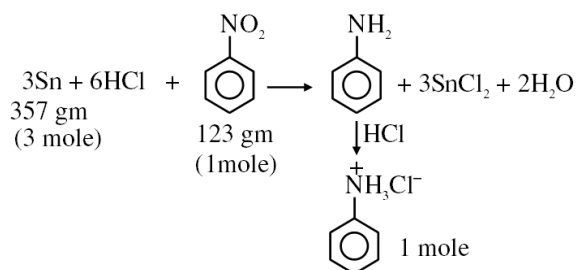
So the amount of nitrobenzene

$$= 0.02 \times 123 = 2.46 \text{ gm}$$

So y is 2.46 gm

4. **Ans (2.46)**

The value of y is



$$(72 + 8 + 35) + 14 = 129 \text{ gm (molecular weight of organic salt)}$$

So to get 2.58 gm organic salt.

We have to form 0.02 mole salt.

So 0.02 mole nitrobenzene is required.

0.06 mole Sn is required.

the amount of Sn required

$$= 0.02 \times 119 \times 3 = 7.14 \text{ gm}$$

So x is 7.14 gm

So the amount of nitrobenzene

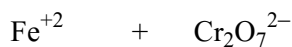
$$= 0.02 \times 123 = 2.46 \text{ gm}$$

So y is 2.46 gm

5. **Ans (2.25)**



x mole x mole



$$\frac{x}{10 \text{ mole}} = \frac{12.5 \text{ ml}}{1000}$$

$$0.03 \text{ M}$$

$$n_f = 1 \quad n_f = 6$$

$$\frac{x}{10} = \frac{12.5 \times 0.03 \times 6}{1000}$$

$$x = 0.0225 \text{ (} x = 2.25 \text{)}$$

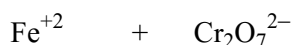
$$\text{wt of Fe} = 1.26 \text{ g}$$

$$\% \text{ Fe} = \frac{1.26}{5.6} \times 100 = 22.5$$

6. **Ans (22.50)**



x mole x mole



$$\frac{x}{10 \text{ mole}} = \frac{12.5 \text{ ml}}{1000}$$

$$0.03 \text{ M}$$

$$n_f = 1 \quad n_f = 6$$

$$\frac{x}{10} = \frac{12.5 \times 0.03 \times 6}{1000}$$

$$x = 0.0225 \text{ (} x = 2.25 \text{)}$$

$$\text{wt of Fe} = 1.26 \text{ g}$$

$$\% \text{ Fe} = \frac{1.26}{5.6} \times 100 = 22.5$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (900)**

$$W_{II} = W_I = 1800 \text{ R} = -1 \times R \times 180 \ln \left(\frac{V_3}{V_2} \right)$$

$$\Delta U_I = nC_{V,m} \Delta T = W_I \{q_I = 0\}$$

$$1800 \text{ R} = 1 \times \frac{5R}{2} \times \Delta T = \Delta T = 720 \text{ K}$$

$$T_2 = 900 \text{ K}$$

8. Ans (33)

$$\lambda = \frac{h}{p} \Rightarrow p = \frac{6.6 \times 10^{-34}}{330 \times 10^{-9}}$$

$$= \frac{4 \times 10^{-3}}{6 \times 10^{23}} \times v \quad (p = m \times v)$$

$$v = 0.33 \text{ m/s} = 33 \text{ cm/s}$$

9. Ans (11)

$$x = +5$$

$$y = +6$$

$$x \text{ is } \text{HIO}_3$$

$$y = \text{Cl}_2\text{O}_6$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C)

$$(A) n_1 = 9 \times 9 \times 9 = 729$$

(B) As per given condition $1 \leq i < j + 2 \leq 10$

$$\Rightarrow j \leq 8 \text{ \& } i \geq 1$$

for $i = 1, 2, \quad j = 1, 2, 3, \dots, 8 \rightarrow (8 + 8)$

possibilities

for $i = 3, \quad j = 2, 3, \dots, 8 \rightarrow 7$ possibilities

$i = 4, \quad j = 3, \dots, 8 \rightarrow 6$ possibilities

$i = 9, \quad j = 1 \rightarrow 1$ possibility

$$\text{So } n_2 = (1 + 2 + 3 + \dots + 8) + 8 = 44$$

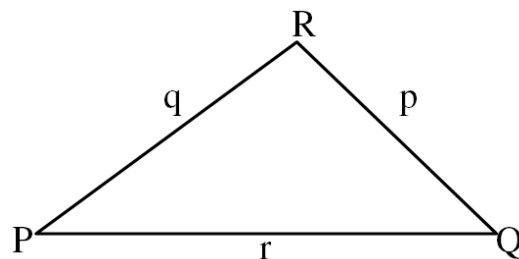
$$(C) n_3 = {}^9C_4 \text{ (Choose any four)}$$

$$= 126$$

$$(D) n_4 = {}^9C_4 \cdot 4! = (126) (24) \Rightarrow \frac{n_4}{12} = 252$$

So correct Ans. (B), (C)

2. Ans (C)



$$(A) \cos P = \frac{q^2 + r^2 - p^2}{2qr}$$

$$= \frac{q^2 + r^2}{2qr} - \frac{p^2}{2qr} \geq 1 - \frac{p^2}{2qr}$$

(as $p^2 + q^2 \geq 2qr$ (AM $\geq$ GM)), so (A) is

incorrect

$$(B) (p + q) \cos R \geq (q - r) \cos P + (p - r) \cos Q$$

$$\Rightarrow (p \cos R + r \cos P) + (q \cos R + r \cos Q) \geq$$

$$q \cos P + p \cos Q$$

$$\Rightarrow q + p \geq r$$

So (B) is incorrect

$$(C) \frac{q+r}{p} = \frac{\sin Q + \sin R}{\sin P} \geq \frac{2\sqrt{\sin Q \times \sin R}}{\sin P}$$

so (C) is correct

$$(D) \cos Q > \frac{p}{r} \Rightarrow \sin R \cos Q > \sin P$$

$$\Rightarrow \sin P + \sin (R - Q) > 2 \sin P$$

$$\Rightarrow \sin (R - Q) > \sin P$$

need not necessarily hold true if $R < Q$

Hence (C)

3. Ans (A,B,D)

(A) Let $g(x) = f(x) - 3 \cos 3x$

Now

$$\int_0^{\pi/3} g(x) dx = \int_0^{\pi/3} f(x) dx - 3 \int_0^{\pi/3} \cos 3x dx = 0$$

Hence $g(x) = 0$ has a root in $(0, \frac{\pi}{3})$

(B) Let $h(x) = f(x) - 3\sin 3x + \frac{6}{\pi}$

Now $\int_0^{\pi/3} h(x) dx$

$$= \int_0^{\pi/3} f(x) dx - 3 \int_0^{\pi/3} \sin 3x dx + \int_0^{\pi/3} \frac{6}{\pi} dx$$

$$= 0 - 2 + 2 = 0$$

Hence $h(x) = 0$ has a root in $(0, \frac{\pi}{3})$

(C) $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{1 - e^{x^2}} = \lim_{x \rightarrow 0} \underbrace{\left(\frac{x^2}{1 - e^{x^2}} \right)}_{-1} \underbrace{\frac{\int_0^x f(t) dt}{x}}_{\text{Apply L' Hopital's Rule}}$

$$= -1 \lim_{x \rightarrow 0} \frac{f(x)}{1} = -1$$

(D) $\lim_{x \rightarrow 0} \frac{\ln(1+x) \int_0^x f(t) dt}{x^2}$

$$= \lim_{x \rightarrow 0} \underbrace{\left(\frac{\ln(1+x)}{x} \right)}_1 \underbrace{\frac{\int_0^x f(t) dt}{x}}_{\text{Apply L' Hopital's Rule}}$$

$$= 1 \lim_{x \rightarrow 0} \frac{f(x)}{1} = 1$$

4. Ans (B,D)

Integrating factor = $e^{\alpha x}$

$$\text{So } ye^{\alpha x} = \int xe^{(\alpha+\beta)x} dx$$

Case-I

If $\alpha + \beta = 0$ $ye^{\alpha x} = \frac{x^2}{2} + c$

It passes through $(1, 2) \Rightarrow C = 2e^{\alpha} - \frac{1}{2}$

$$\text{So } ye^{\alpha x} = \frac{x^2 - 1}{2} + 2e^{\alpha}$$

for $\alpha = 1$

$$y = \frac{x^2}{2} e^{-x} + \left(2e - \frac{1}{2} \right) e^{-x} \rightarrow (B)$$

Case-II

If $\alpha + \beta \neq 0$

$$ye^{\alpha x} = \frac{x \cdot e^{(\alpha+\beta)x}}{\alpha + \beta} - \frac{1}{\alpha + \beta} e^{(\alpha+\beta)x} dx$$

$$\Rightarrow ye^{\alpha x} = \frac{xe^{(\alpha+\beta)x}}{\alpha + \beta} - \frac{e^{(\alpha+\beta)x}}{(\alpha + \beta)^2} + c$$

$$\Rightarrow \text{So } c = 2e^{\alpha} - \frac{e^{\alpha+\beta}}{\alpha + \beta} + \frac{e^{\alpha+\beta}}{(\alpha + \beta)^2}$$

$$y = \frac{e^{\beta x}}{(\alpha + \beta)^2} ((\alpha + \beta)x - 1) + e^{-\alpha x}$$

$$\left(2e^{\alpha} - \frac{e^{\alpha+\beta}}{\alpha + \beta} + \frac{e^{\alpha+\beta}}{(\alpha + \beta)^2} \right)$$

If $\alpha = \beta = 1$

$$y = \frac{e^x}{4} (2x - 1) + e^{-x} \left(2e - \frac{e^2}{2} + \frac{e^2}{4} \right)$$

$$y = \frac{e^x}{2} \left(x - \frac{1}{2} \right) + e^{-x} \left(2e - \frac{e^2}{4} \right) \rightarrow (D)$$

5. **Ans (A,D)**

$$\vec{OB} \times \vec{OC} = \frac{1}{2} \vec{OB} \times (\vec{OB} - \lambda \vec{OA})$$

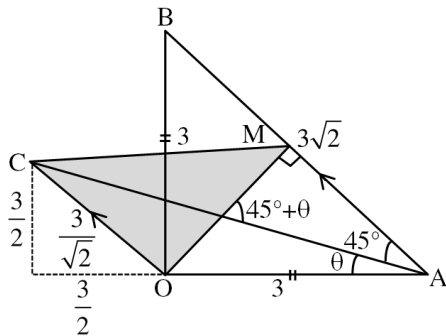
$$= \frac{\lambda}{2} (\vec{OA} \times \vec{OB})$$

$$|\vec{OB}| \times |\vec{OC}| = \frac{|\lambda|}{2} |\vec{OA}| \times |\vec{OB}|$$

(Note $\vec{OA}$ & $\vec{OB}$ are perpendicular)

$$\Rightarrow \frac{9\lambda}{2} = \frac{9}{2} \Rightarrow \lambda = 1 \text{ (given } \lambda > 0)$$

$$\text{So } \vec{OC} = \frac{\vec{OB} - \vec{OA}}{2} = \frac{\vec{AB}}{2}$$



M is mid point of AB

Note projection of $\vec{OC}$ on $\vec{OA} = -\frac{3}{2}$

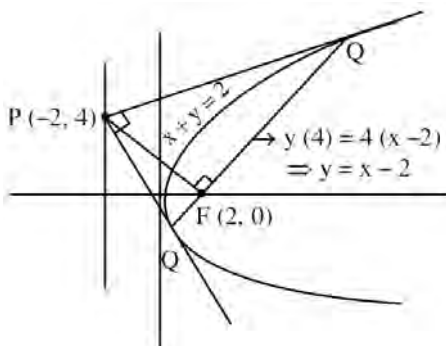
$$\tan \theta = \frac{1}{3}$$

$$\text{Area of } \triangle ABC = \frac{9}{2}$$

Acute angle between diagonals is

$$\tan^{-1} \left(\frac{1 + \frac{1}{3}}{1 - \frac{1}{3}} \right) = \tan^{-1} 2$$

6. **Ans (B,C,D)**



Note that P lies on directrix so triangle PQQ' is right angled, hence QQ' passes through focus F.
PF = $4\sqrt{2}$

Equation of QF is $y = x - 2$ & PF is $x + y = 2$
 $\Rightarrow PF \perp QF \Rightarrow \triangle PFQ$ is a right angle triangle
Hence B, C, D.

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A)**

$$S_n = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{n-1}}$$

$$= 2 \left(1 - \frac{1}{2^n} \right) = 2 - \frac{1}{2^{n-1}}$$

$$\text{Centre of } C_n \text{ is } \left(2 - \frac{1}{2^{n-2}}, 0 \right)$$

$$\text{and radius of } C_n \text{ is } \frac{1}{2^{n-1}}$$

$$\text{when } r = \frac{1025}{513} < 2$$

C_n will lie inside M

$$\text{when } 2 - \frac{1}{2^{n-2}} + \frac{1}{2^{n-1}} < \frac{1025}{513}$$

$$\Rightarrow k = 10$$

$$\text{Also } \ell = 5$$

$$3k + 2\ell = 30 + 10 = 40$$

8. **Ans (D)**

Center of D_n is (S_{n-1}, S_{n-1})

$$r = \frac{1}{2^{n-1}}$$

D_n will lie inside

$$\text{when } \sqrt{2} (S_{n-1}) < \left(\frac{2^{199} - 1}{2^{198}} \right) \sqrt{2} - \frac{1}{2^{n-1}}$$

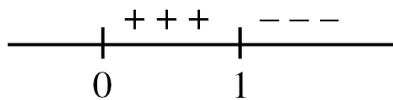
$$\Rightarrow \frac{\sqrt{2}}{2^{n-2}} > \frac{\sqrt{2}}{2^{198}} + \frac{1}{2^{n-1}}$$

$$\Rightarrow n = 199$$

9. Ans (D)

$$f'(x) = (|x| - x^2)e^{-x^2} + (|x| - x^2)e^{-x^2}, \quad x \geq 0$$

$$f' = 2(x - x^2)e^{-x^2} \dots (1)$$



hence option (D) is correct

$$g'(x) = xe^{-x^2} 2x$$

$$g'(x) = 2x^2e^{-x^2} \dots (2)$$

$$\Rightarrow g'(x) \geq 0 \quad \forall x \in \mathbb{R}$$

hence (C) is wrong

by (1) + (2)

$$f'(x) + g'(x) = 2 \times e^{-x^2}$$

$$f(x) + g(x) = -e^{-x^2} + c$$

$$f(x) + g(x) = -e^{-x^2} + 1 \quad \text{as } f(0) = g(0) = 0$$

$$F(\sqrt{\ln 3}) + g(\sqrt{\ln 3}) = 1 - \frac{1}{3} = \frac{2}{3} \quad (\text{option})$$

(A) is wrong

$$H(x) = \psi_1(x) - 1 - \alpha x = e^{-x} + x - 1 - \alpha x,$$

$$x \geq 1 \text{ \& } \alpha \in (1, x)$$

$$H(1) = e^{-1} + 1 - 1 - \alpha < 0$$

$$H'(x) = -e^{-x} + 1 - \alpha > 0$$

$\Rightarrow H(x)$ is $\downarrow \Rightarrow$ option (B) is wrong

10. Ans (B)

$$(A) \psi_1(x) = e^{-x} + x, \quad x \geq 0$$

$$\psi_1'(x) = 1 - e^{-x} > 0 \Rightarrow \psi_1(x) \text{ is } \uparrow$$

$$\psi_1(x) \geq \psi_1(0) \quad \forall x \geq 0 \Rightarrow \psi_1(x) \geq 1$$

$$(B) \psi_2(x) = x^2 - 2x + 2 - 2e^{-x} \quad x \geq 0$$

$$\psi_2'(x) = 2x - 2 + 2e^{-x} = 2\psi_1(x) - 2 \geq 0 \quad \forall x \geq 0$$

$$\Rightarrow \psi_2(x) \text{ is } \uparrow \Rightarrow \psi_2(x) \geq \psi_2(0) \Rightarrow \psi_2(x) \geq 0$$

$$(C) f(x) = 2 \int_0^x (t - t^2)e^{-t^2} dt \text{ \& } x \in \left(0, \frac{1}{2}\right)$$

$$= \int_0^x 2te^{-t^2} dt - \int_0^x 2t^2e^{-t^2} dt$$

$$= -e^{-x^2} \Big|_0^x -$$

Let $H(x)$

$$= f(x) - 1 + e^{-x^2} + \frac{2}{3}x^3 - \frac{2}{5}x^5, \quad x \in \left(0, \frac{1}{2}\right)$$

$$H(0) = 0$$

$$H'(x) = 2(x - x^2)e^{-x^2} - 2xe^{-x^2} + 2x^2 - 2x^4$$

$$= -2x^2e^{-x^2} + 2x^2 - 2x^4$$

$$= 2x^2 \left(1 - x^2 - e^{-x^2}\right)$$

$$\therefore e^{-x} \geq 1 - x \quad \forall x \geq 0$$

$$\Rightarrow H'(x) \leq 0$$

$$\Rightarrow H(x) \text{ is } \downarrow \Rightarrow 1 - 1(x) < 0 \quad \forall x \in \left(0, \frac{1}{2}\right)$$

$$\text{Let } P(x) = g(x) - \frac{2}{3}x^3 + \frac{2}{5}x^5 - \frac{1}{7}x^7 \in \left(0, \frac{1}{2}\right)$$

$$P'(x) = 2x^2e^{-x^2} - 2x^2 + 2x^4 - x^6$$

$$= 2x^2 \left(1 - \frac{x^2}{1} + \frac{x^4}{2} - \frac{x^6}{3} + \dots\right) - 2x^2 + 2x^4 - x^6$$

$$= -\frac{x^8}{3} + \frac{x^{10}}{12} \dots \dots \dots$$

$$\Rightarrow P'(x) \leq 0$$

$$\Rightarrow P(x) \text{ is } \downarrow$$

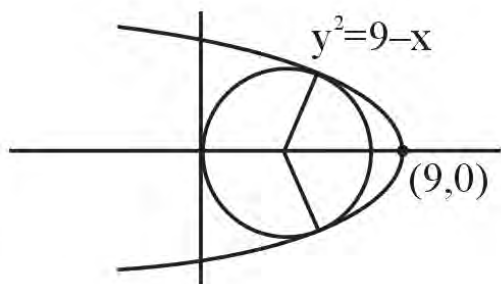
$$\Rightarrow P(x) \leq 0$$

option (B) is correct

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (2.50)**



Let the circle be

$$x^2 + y^2 + \lambda x = 0$$

For point of intersection of circle & parabola

$$y^2 = 9 - x.$$

$$x^2 + 9 - x + \lambda x = 0 \Rightarrow x^2 + x(\lambda - 1) + 9 = 0$$

$$\text{For tangency : } \Delta = 0 \Rightarrow (\lambda - 1)^2 - 36 = 0$$

$$\Rightarrow \lambda = 7 \text{ (rejected) or } \lambda = -5$$

$$\text{Circle : } x^2 + y^2 - 5x = 0$$

$$\text{Radius} = \frac{5}{2} = 2.5$$

2. **Ans (3.00)**

For point of intersection :

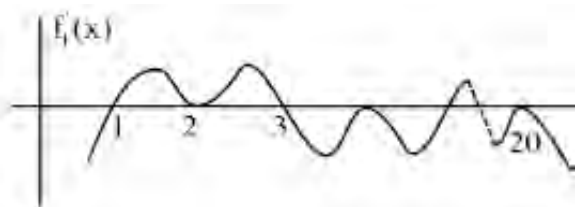
$$x^2 - 6x + 9 = 0 \Rightarrow x = 3 \text{ so } \alpha = 3$$

3. **Ans (50.00)**

$$f_1(x) = \int_0^x \prod_{j=1}^{20} (t - j)^j dt$$

$$f_1'(x) = \prod_{j=1}^{20} (x - j)^j$$

$$= (x - 1)(x - 2)^2(x - 3)^3 \dots (x - 20)^{20}$$



So points of minima are $4m + 1$ where

$$m = 0, 1, \dots, 4 \Rightarrow m_1 = 5$$

Points of maxima are $4m - 1$ where

$$m = 1, 2, \dots, 5 \Rightarrow n_1 = 5$$

$$\Rightarrow 2m_1 + 3n_1 + m_1n_1 = 50$$

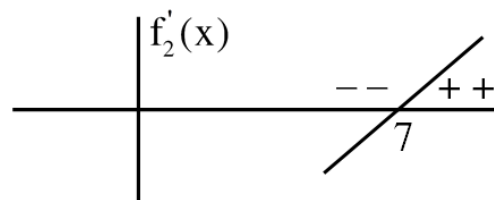
4. **Ans (4.00)**

$$f_2'(x) = 98(x - 1)^{50} - 600(x - 1)^{49} + 2450$$

$$\Rightarrow f_2'(x) = 2 \times 49 \times 50(x - 1)^{49} - 50 \times 12 \times 49(x - 1)^{48}$$

$$= 50 \times 49 \times 2(x - 1)^{48}(x - 1 - 6)$$

$$= 50 \times 49 \times 2(x - 1)^{48}(x - 7)$$



Point of minima = 7

$$\Rightarrow m_2 = 1$$

No point of maxima

$$\Rightarrow n_2 = 0$$

$$4m_2 + 6n_2 + 8m_2n_2 = 4$$

5. **Ans (3.00)**

$$S_1 = \int_{\pi/16}^{7\pi/16} f(x) dx = \int_{\pi/16}^{7\pi/16} \sin^2 x dx \quad \dots (1)$$

$$S_1 = \int_{\pi/16}^{7\pi/16} \sin^2 \left(\frac{\pi}{16} + \frac{7\pi}{16} - x \right) dx$$

$$= \int_{\pi/16}^{7\pi/16} \cos^2 x dx \quad \dots (2)$$

by (1) + (2)

$$2S_1 = \int_{\pi/16}^{7\pi/16} (\sin^2 x + \cos^2 x) dx$$

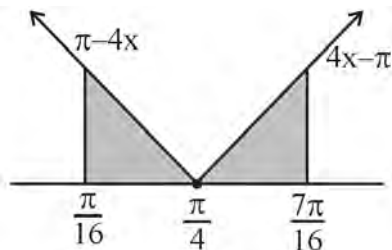
$$2S_1 = \int_{\pi/16}^{7\pi/16} dx$$

$$2S_1 = \frac{7\pi}{16} - \frac{\pi}{16} = \frac{3\pi}{8}$$

$$S_1 = \frac{3\pi}{16}$$

$$\Rightarrow \frac{16S_1}{\pi} = 3$$

6. Ans (16.50)



$$S_2 = \int_{\pi/16}^{7\pi/16} f(x) g_2(x) dx = \int_{\pi/16}^{7\pi/16} \sin^2 x |4x - \pi| dx \quad \dots (1)$$

$$= \int_{\pi/16}^{7\pi/16} \sin^2 \left(\frac{\pi}{2} - x \right) \left| 4 \left(\frac{\pi}{2} - x \right) - \pi \right| dx$$

$$S_2 = \int_{\pi/16}^{7\pi/16} (\cos^2 x) |\pi - 4x| dx \quad \dots (2)$$

by (1) + (2)

$$\Rightarrow 2S_2 = \int_{\pi/16}^{7\pi/16} |4x - \pi| (\sin^2 x + \cos^2 x) dx =$$

$$\int_{\pi/16}^{7\pi/16} |4x - \pi| dx$$

$$2S_2 = \frac{33\pi^2}{64}$$

$$\frac{64S_2}{\pi^2} = \frac{33}{2} = 16.5$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (2140)

A = set of numbers divisible by 3

A = {3, 6, 9, 12, 999}

$\therefore n(A) = 333$

B = set of numbers divisible by 7

B = {7, 14, 21, 994}

$\therefore n(B) = 142$

$A \cap B = \{21, 42, \dots, 987\}$

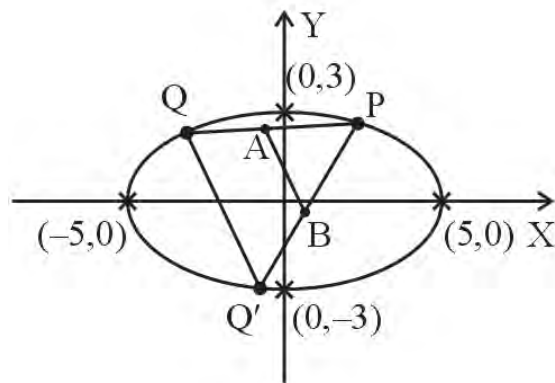
$\Rightarrow n(A \cap B) = 47$

$\therefore n(A \cup B) = 333 + 142 - 47 = 428$

required probability = $\frac{428}{1000} = P$

so, $5000 P = \frac{428}{1000} \times 5000 = 2140$

8. Ans (5)



A and B be midpoints of segment PQ and PQ' respectively

AB = distance between M(P, Q) and M(P, Q')

$$= \frac{1}{2} \cdot QQ'$$

Since, Q, Q' must be on E, so,

maximum of QQ' = 10

$$\therefore \text{Maximum of AB} = \frac{10}{2} = 5$$

9. Ans (155)

$$\text{Let } f(x) = \left(\frac{10x}{x+1} \right)$$

$$\text{So, } f'(x) = 10 \left(\frac{(x+1) - x}{(x+1)^2} \right) = \frac{10}{(x+1)^2} > 0$$

$\forall x \in [0, 9]$,

So, $f(x)$ is an increasing function

So, range of $f(x)$ is $[0, 9]$

$$I = \int_0^{1/9} \left[\sqrt{\frac{10x}{x+1}} \right] dx + \int_{1/9}^{2/3} \left[\sqrt{\frac{10x}{x+1}} \right] dx + \int_{2/3}^9 \left[\sqrt{\frac{10x}{x+1}} \right] dx$$

$$= 0 + \int_{1/9}^{2/3} dx + 2 \int_{2/3}^9 dx$$

$$= \frac{2}{3} - \frac{1}{9} + 2 \left(9 - \frac{2}{3} \right)$$

$$= \frac{6-1}{9} + 2 \times \frac{25}{3} = \frac{5}{9} + \frac{50}{3}$$

$$I = \frac{5+150}{9} = \frac{155}{9}$$

$$\Rightarrow 9I = 155$$